safety series

safety series

No.88

RECOMMENDATIONS

Medical Handling of Accidentally Exposed Individuals

INTERNATIONAL ATOMIC ENERGY AGENCY, VIENNA, 1988

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MEDICAL HANDLING OF ACCIDENTALLY EXPOSED INDIVIDUALS

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FOREWORD

In 1978, the International Atomic Energy Agency issued a publication in the IAEA Safety Series entitled Manual on Early Medical Treatment of Possible Radiation Injury (Safety Series No. 47). The contents were directed to first aid and early medical treatment of workers who might be concerned in an accident involving exposure to radiation, whether external or internal.

The present manual is the first of a set of three safety series publications dealing with assessment and treatment of overexposures which will constitute an articulated system of documents covering all aspects of diagnosis, prognosis and treatment of overexposures. This document establishes a set of general criteria and recommendations to aid specialists involved in the medical handling of overexposed persons. The other two documents which are still under preparation are of a more pragmatic, operational nature. They can be aptly defined as shelf books or ''how to'' handbooks. They will deal with external irradiation, and internal and external contamination.

Many lessons have been learned from some serious accidents which occurred in the recent past and have caused serious acute health problems to both workers and to members of the public.

The most relevant of these lessons have been incorporated into the present document. Some of them will also be discussed and described in greater detail in the other two technical handbooks, which will complete the set.

The International Atomic Energy Agency asked the following experts to prepare the present manual: Dr. H. Jammet and Dr. J.C. Nenot of the Commissariat à l'énergie atomique, Département de protection, Service de protection sanitaire, Fontenay-aux-Roses, France; Dr. H.T. Daw, Cairo, Egypt, retiree, formerly of IAEA, Division of Nuclear Safety and Radiological Protection; Dr. H.-D. Rödler of the Federal Health Office, Institute for Radiation Hygiene, Federal Republic of Germany; Dr. S. Rae, United Kingdom; and Dr. C. Lushbaugh, Oak Ridge Associated Universities, Oak Ridge, United States of America. Dr. A. Bianco, IAEA staff member, co-ordinated the experts' work. The contribution of Ms. A.-M. Schmitt-Hannig in reviewing the final draft is kindly acknowledged.

This document is intended mainly for physicians who specialize in occupational medicine, industrial hygiene, preventive medicine and public health, and for governmental and regulatory authorities.

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CONTENTS

CHAPTER 1

INTRODUCTION

This report deals with the medical management of individual patients or small groups of patients under close medical supervision. Additional and quite different considerations arise in applying radiological protection principles in the public health field or in the consideration of prophylactic responses to widespread environmental contamination. These further aspects are not considered here but are the subject of other reports by the IAEA and other international and national authorities [1, 2].

1.1. ACCIDENTAL EXPOSURE

During the past few years the use of ionizing radiation and radionuclides has shown a continued increase. The benefits arising from their use are clear-cut but it has to be kept in mind that there is an associated risk to the user or the environment unless proper procedures are followed.

Accidental exposure to external radiation may arise from the use of:

— sealed sources in industry, e.g. for radiography;
in medicine, for therapy, diagnosis and sterilization;
in research, scientific and applied research, e.g. metallurgy;

— X-ray equipment
and accelerators in medicine, for diagnosis and therapy;
in industry, e.g. analytical procedures
in research and industrial applications;

while external and internal contamination may result from the use of:

— unsealed sources in industry, e.g. nuclear power production and its associated fields;
in medicine, e.g. nuclear medical procedures;
in research, in a number of fields.

The purpose of this document is to provide the physician with a knowledge of the principles involved in the management of accidentally exposed individuals.

1.2. TREATMENT OF EXTERNAL IRRADIATION, AND EXTERNAL AND INTERNAL CONTAMINATION

The aim of diagnosis and treatment of accidentally exposed cases is to mitigate the actual and potential radiation damage and the resulting complications. There are, however, substantial differences in approach between the treatment of external irradiation and the treatment of internal contamination cases.

External irradiation may be of varying intensity and may affect various regions of the body. Dosimetric information (i.e. nature and quality of radiation, magnitude of dose, time of its delivery) — if available — is a useful means of deciding which treatment procedures are appropriate and where they should be carried out (e.g. whether the patient should be removed to a specialized centre). However, the treatment, particularly surgery and treatment using invasive and risky methods must take into account the patient's condition, the ensuing signs and symptoms, their intensity and rapidity of occurrence. Clinical judgement should retain priority over dosimetric information, since in accidental situations the latter may be fragmentary and inaccurate.

In cases of internal contamination, it is extremely rare, if ever, that internal irradiation causes early clinically recognizable signs and symptoms. Biological effects, stochastic or non-stochastic, may only occur later. On the other hand, treatment has to be applied immediately after the occurrence of contamination if it is to be effective. Selection of an appropriate treatment procedure depends primarily on the radionuclide in question, its activity, and its physical and chemical form. At the present time, the toxicity of the available decontaminants is usually very low so they may be applied promptly even if the activity of the contaminating radioisotope, and therefore the radiological risk to the patient, might have been overestimated.

Treatment of external contamination should be undertaken as soon as possible and the effectiveness of the treatment assessed by direct measurement of the deposited activity.

1.3. EXTERNAL EXPOSURE

External exposure could involve the whole body, part of the body or be localized (see Chapter 2).

The typical phases of the acute radiation syndrome (ARS) consist of:

1. Prodromal phase — a few hours to several days;
2. Latent period — one to three weeks;
3. Critical phase — in the haematopoietic phase usually beginning at 18 to 21 days. The shorter the onset, the more guarded is the prognosis.

If only part of the body is involved the clinical course is more benign than if the entire body is involved. Since the treatment of prodromal symptoms and signs is non-specific, immediate definitive treatment can be deferred until experienced medical consultation can be obtained. During the first few days consultation with medical radiation physicists and health physicists will be helpful in determining management although in some instances only clinical observation and the use of laboratory tests (see Chapters 2 and 3) will be available.

TABLE I. CLINICAL COURSE OF ACUTE RADIATION SYNDROME

Acute radiation syndrome	Approximate duration (for haematopoietic syndrome)
Time from exposure to symptoms	Minutes to hours
Prodromal phase	1-7 days
Latent phase	7-21 days
Critical phase	From second or third week up to the seventh week
Recovery	8-15 weeks

Symptoms and signs in prodromal stage of acute radiation syndrome

(in approximate order of increasing severity)

Anorexia	Shock
Nausea	Oliguria
Vomiting	Ataxia
Weakness and fatigue	Disorientation
Prostration	Coma
Diarrhoea	Death
Conjunctivitis	
Erythema	
Hyperaesthesia, paraesthesia	

Symptoms and signs in critical phase of acute radiation syndrome

Fever	Infection
Abdominal pain	Purpura
Abdominal distention	Haemorrhage
Scalp pain	Weight loss
Epilation	

The careful recording of thorough clinical examinations performed daily is essential. The use of photographs and tape recording is suggested. Table I provides an outline to record important changes. These observations should be carefully co-ordinated with laboratory tests.

Daily routine haematological tests are essential. In the first one to two days absolute lymphocyte counts and granulocyte counts, perhaps as often as two to three each day, will be especially valuable in highly exposed (clinically very ill) patients. Samples for chromosome analysis should be obtained early.

In the event that several persons or large numbers are involved, sorting (triage) must be considered to direct relatively limited clinical facilities and personnel as effectively as possible. Several laboratory tests — the absolute lymphocyte count, the granulocyte count — will be found to be especially useful (see Chapters 2 and 3). If the site of the accident is such that neutron exposure is a consideration, blood must be taken for ^{24}Na measurement.

Recent observations suggested that the onset of nausea and vomiting within the first 12–24 hours after exposure is likely to be due to radiation. If the occurrence is later its cause may be due to emotional factors.

Radioactive contamination — Although contamination is greatly feared by most people, it is essential to recall that there are no reported cases in which contamination has produced prodromal symptoms. Furthermore, early or late non-stochastic effects are rare except in very unusual circumstances [3]. Thus although decontamination is more effective at early times, one must remember that the effects of vigorous therapy must be balanced against its concomitant and later risks versus benefits. The major goal of decontamination is to prevent late effects, notably cancer that in most instances will not develop before five to ten years.

For localized radiation injury the treatment is essentially directed to the management of the local injury (see Section 3.6.3).

CHAPTER 2

ACCIDENTAL EXTERNAL EXPOSURE

2.1. INTRODUCTION

Accidental external exposure may result from photon irradiation (X-rays or gamma), particle irradiation (electrons, protons, deuterons, heavy ions) or from mixed irradiation (gamma and neutrons).

The exposure may involve the whole body or may be confined to a sizeable part of the body, and if the exposure is severe it results in ARS. The exposure may also be localized, from narrow beam irradiation or a small radiation source held accidentally, for example, in a hand.

2.2. WHOLE BODY EXPOSURE

2.2.1. Acute exposure

2.2.1.1. Dose dependence of effects

The seriousness of an accident depends on the absorbed dose and its distribution in the body. In the case of a single whole body irradiation delivered in a short period of time the patient's survival is threatened at doses of 2 Gy or more. In the absence of treatment or in the case of serious injury it is estimated that after a total body dose of about 3.5 Gy the mortality after two months will be 50%.

Depending upon the whole body dose, four forms of the acute radiation syndrome can be distinguished:

(a) the haematopoietic, resulting from doses in the range 1–10 Gy;
(b) the gastrointestinal (10–20 Gy);
(c) the cardiovascular or toxaemic (20–50 Gy);
(d) the nervous system syndrome (above 50 Gy).

The severity of the clinical features of the haematopoietic syndrome resulting from the irradiation may be graded as follows:

— From 0 to 0.25 Gy: No clinical symptoms but a slightly increased frequency of chromosome aberrations may be detected in lymphocytes.
— From 0.25 to 1 Gy: Either no symptoms or transient nausea. Biological tests may reveal a lymphopenia accompanied in some cases by slight thrombopenia. Cytogenetic changes in lymphocytes are readily detected. Some studies have shown slight changes in the patient's electroencephalogram.

TABLE II. SUMMARY OF CLINICAL FINDINGS, THERAPY AND PROGNOSIS IN MAN AFTER WHOLE BODY IRRADIATION[a] (provided by E.L. Saenger, December 1987)

	Injury groups						
	0	I	II	III	IV	V	VI
USSR classification (degree)		1st	2nd	3rd	4th		
Approximate dose (Gy)	0.25–0.75	0.5–2	2.1–4	4.1–6	6.1–15	15.1–50	50
Time of onset of prodromes, duration	0	2–24 h	1–48 h	0.5–1 h	0.5 h	0.–0.5 h	immediate
Frequency of nausea and vomiting (%)	0	20–70	10–80	90–100	100	100	100
Important findings	no clinical	minimal prodromata, mild lymphopenia, 600–1000	mild prodromata, mild blood cell depression, infection	serious course, severe marrow depression, early gastroenteric damage, infection, haemorrhage	accelerated course of ARS, severe GI changes, >8 Gy	fulminating course with cardiac and/or CNS impairment	no real human data, CNS changes to immediate death
Skin involvement	0	slight	slight	severe, depending on contamination	immediate erythema and contamination effects	severe erythema	—

Injury groups

	0	I	II	III	IV	V	VI
USSR classification (degree)		1st	2nd	3rd	4th		
Approximate dose (Gy)	0.25–0.75	0.5–2	2.1–4	4.1–6	6.1–15	15.1–50	50
Latent period	—	20–30 d	15–25 d	8–17 d	6–8 d	0–2 d	0
Duration of critical phase	—	3–6 w	12–15 w	1–26 w	prolonged	—	—
Time to recovery	3–6 w	6–10 w	6–12 m	0.5–15 m	15–24 m	—	—
Prognosis	excellent	excellent	guarded	guarded	poor	—	—
Lethality	0	0	0–50%	probably	most likely	100%	100%
Time of death	—	—	1–3 m	2–8 w	10–50 d	0.5–14 d	1–2 d
Cause of death	—	—	haemorrhage, infection	haemorrhage, infection	gastroenteric and haematological failure	circulatory collapse, cerebral oedema	cerebral necrosis(?)

TABLE II. (cont.)

	Injury groups						
	0	I	II	III	IV	V	VI
USSR classification (degree)		1st	2nd	3rd	4th		
Approximate dose (Gy)	0.25–0.75	0.5–2	2.1–4	4.1–6	6.1–15	15.1–50	50
Absolute lymphocytes/μL (1–3 d)	1500–3000	600–1000	300–600	100–500	< 100	—	—
Absolute granulocytes/μL and nadir	—	1500–2000	> 600–1500	< 1000 (8–20 d)	< 500 (7–9 d)	—	—
Absolute platelets/μL and nadir	—	40 000–60 000 (25–28 d)	40 000 (17–24 d)	< 40 000 (10–16 d)	< 40 000 (8–10 d)	—	—

[a] Values in this table should be regarded as approximate and may vary in either direction by at least 25%.

Normally only symptomatic treatment is given but medical surveillance should be continued for several days.

— 1–2 Gy: Mild degree of the haematopoietic form; nausea and vomiting occur in only a fraction of the irradiated persons. The symptoms become manifest in the first hours after exposure. Development of neutropenia and thrombopenia takes four to six weeks. The degree of the cytopenia does not frequently lead to bleeding and infection. The majority of the patients recover without treatment; however, to prevent infection and haemorrhagic symptoms, careful haematological follow-up and care are required.
— 2–4 Gy: Moderate degree of the haematological syndrome. Nausea and vomiting appear after one to two hours and in most victims, the nadir of neutro- and thrombopenia develops in three to four weeks. In all patients these symptoms are accompanied by fever and bleeding. Under contemporary therapeutic conditions all patients are likely to recover.
— 4–6 Gy: Severe syndrome; nausea and vomiting appear 0.5–1 hour after irradiation. Other prodromal symptoms are manifest: early fever, erythema of skin and mucosae. The deepest fall in neutrophils and thrombocytes occurs at two to three weeks and very low values are reached. The latter are maintained for a period of about two weeks. Without therapy, the majority of the patients will die of infections and consequences of bleeding. If adequate supportive therapy is instituted the majority are likely to recover.
— 6–10 Gy: Extremely severe haematopoietic syndrome. Nausea and vomiting appear very early, less than 30 minutes after the exposure. In a substantial proportion of the victims, diarrhoea occurs one to two hours after the irradiation. Maximal cytopenia appears at 10–14 days; moreover, as a rule, at six to eight days acute radiation stomatitis becomes manifest. Sometimes at eight to ten days radiation induced enteritis is seen. Without adequate therapy the lethality reaches 100%. If appropriate therapy is started early, a fraction of the patients may still recover. The lethality results from the coincidence of severe impairment of the haematopoietic function and radiation injury to other organs, mostly of the pharyngeal mucosa, oesophagus and intestines. Injury to pulmonary tissue may also play a significant role.

Additional information about the time dependence of appearance and severity of clinical signs and symptoms as related to the absorbed whole body dose is given in Tables II and III.

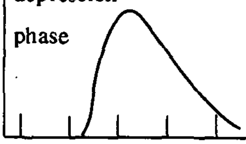
2.2.1.2. The acute radiation syndrome

The clinical picture following substantial whole body acute exposure is well documented. Table I summarizes the symptoms and clinical signs which are typical for the acute radiation syndrome (ARS). A schematic and theoretical representation

TABLE III. ACUTE RADIATION SYNDROME FEATURES[a]
(from Radiation Injury, by A.C. Upton, Chicago (1969))

Salient features of the haematopoietic form of the acute radiation syndrome in man

Severity of reaction

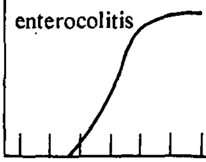
Haematopoietic form	Characteristic signs and symptoms	Underlying pathology
Prodromal phase 	Anorexia Nausea Vomiting	
Bone marrow depression phase 	Purpura Haemorrhage Susceptibility to infection	Bone marrow aplasia Leukopenia Thrombocytopenia Anaemia

0 2 4 6 8

Weeks post-exposure

Salient features of the gastrointestinal form of the acute radiation syndrome

Severity of reaction

Gastrointestinal form	Characteristic signs and symptoms	Underlying pathology
Prodromal phase 	Anorexia Nausea Vomiting	
Radiation induced enterocolitis 	Fever Diarrhoea Disturbance of electrolyte balance	Mitotic arrest in epithelium Denudation and ulceration of intestinal wall

0 2 4 6 8 20 12

Days post-exposure

TABLE III. (cont.)

Central nervous system form of the acute radiation syndrome

Cerebral form	Characteristic signs and symptoms	Underlying pathology
Prodromal phase	Anorexia Nausea Vomiting	
Inflammatory reactions	Apathy Lethargy Somnolence	Vasculitis Meningitis Encephalitis Brain oedema
Convulsive phase	Tremor Convulsions Ataxia	Pyknosis of cerebellar granule cells

Severity of reaction

0 2 4 6 8 20 12

Hours post-exposure

[a] Cardiovascular and toxemic syndromes are not included in this table.

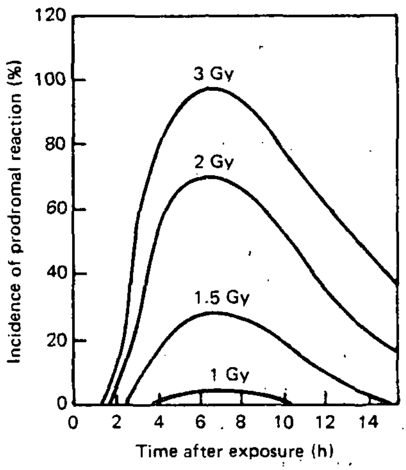


FIG. 1. Estimated incidence and timing of prodromal symptoms in man in relation to absorbed doses received at a low linear energy transfer (LET).

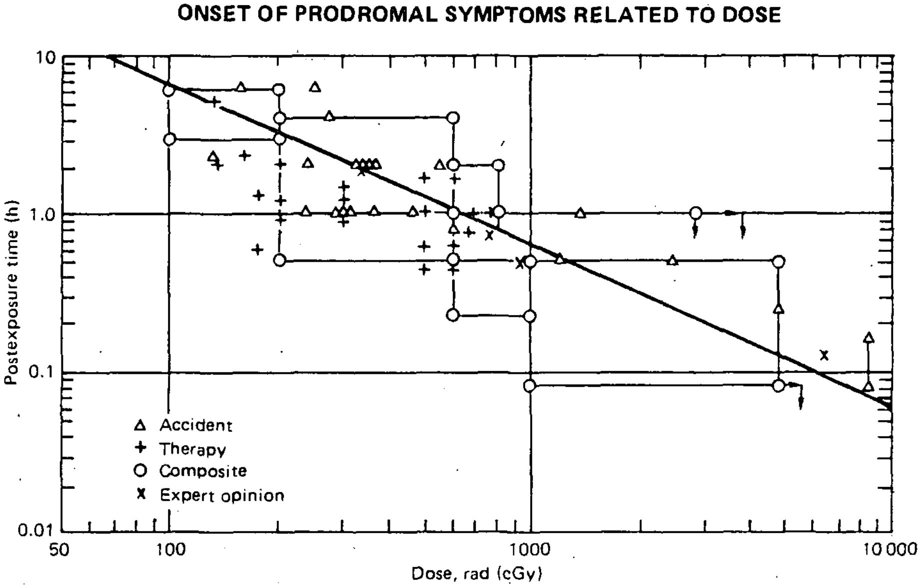


FIG. 2. The course of events in human cases (from S.J. Baum et al. [4]).

of the frequency of the prodromal reaction as a function of absorbed dose and time after exposure is given in Fig. 1.

Initial symptoms may serve as a reasonable guide for the assessment of the exposure severity. At doses below or near those corresponding to the median lethal range, the main symptoms are anorexia, nausea and vomiting. The shorter the interval between exposure and the onset of nausea and vomiting, the more severe these symptoms become and the longer their continuance the more serious the prognosis. However, psychosomatic symptoms produced by fear, or from witnessing the distress of heavily exposed persons, may complicate this picture. Early neurological symptoms such as apathy, ataxia or convulsions indicate a very high dose to the whole body or to the central nervous system.

Figure 2 shows the course of events in human cases.

Early erythema and conjunctivitis may be useful indicators of the spatial distribution of the dose and its magnitude. The latent period of erythema and the magnitude of the dose vary inversely, and the skin should therefore be examined frequently. Colour photographs of the development of skin changes should be obtained so that their evolution may be followed.

If the dose received by the whole body is assessed as being very high or if the irradiation was directed mainly towards the head, a neurological examination should be made and, if possible, an electroencephalogram arranged. In the case of possible high dose to the thorax, a cardiovascular examination, including an electrocardiogram, may be indicated.

The dose dependence of the symptoms and the prognosis following acute whole body irradiation are summarized in Table II. This table also identifies the three classical forms of the acute radiation syndrome: the haematopoietic form in the dose range of 2–10 Gy, the gastrointestinal form (GI) in the range of 10–15 Gy and the central nervous system (CNS) form in the range above 50 Gy. The severity of the reaction and its relation to the time following exposure, the clinical factors and the underlying pathology are detailed in Table III.

2.2.1.3. Biological indicators of radiation exposure

The severity of injury can be assessed by using certain biological indicators. These include:

(i) *Changes in the peripheral blood.* The changes in peripheral blood counts following whole body exposure may serve as biological indicators of the magnitude of the dose. Depletion of the lymphocytes in peripheral blood is one such indicator. The dependence of the rate of decrease of the count in the first days post-exposure upon the absorbed dose in the whole body is presented in Figs 3 and 4 [5].

(ii) *Chromosome aberrations in peripheral blood lymphocytes.* This is a very useful indicator of the absorbed dose. However, results cannot become available earlier than 2.5–3 days after sampling. The method itself is laborious and time consuming, even if partial automation in recent times has somewhat facilitated its application.

In quantitative terms the method is particularly useful when relatively uniform whole body irradiation has taken place. In cases of substantive non-uniformity of dose distribution the estimates of mean whole body dose may be inaccurate. For irradiations of a very limited proportion of the body, the intercellular distribution of asymmetric chromosomal exchanges (dicentrics, rings) differs from that typical for uniform irradiation and, therefore, may provide useful information on characteristics of the exposure. However, minor deviations from the uniformity of dose distribution and consequent sparing of lymphocyte forming stem cells, even if prognostically important, cannot be detected by this type of analysis. Blood for lymphocyte culture should be sampled as early as possible and results of scoring compared with an appropriate reference curve, established from in vitro irradiation of whole blood with a corresponding radiation at standardized conditions (Fig. 5). With a reasonable effort, doses down to 0.1 Gy can be detected, but at the mid-lethal doses the method proved particularly useful for verification of doses in the range. At doses above 1 Gy, the accuracy of the method enables prognostic inferences to be drawn from results of the analysis.

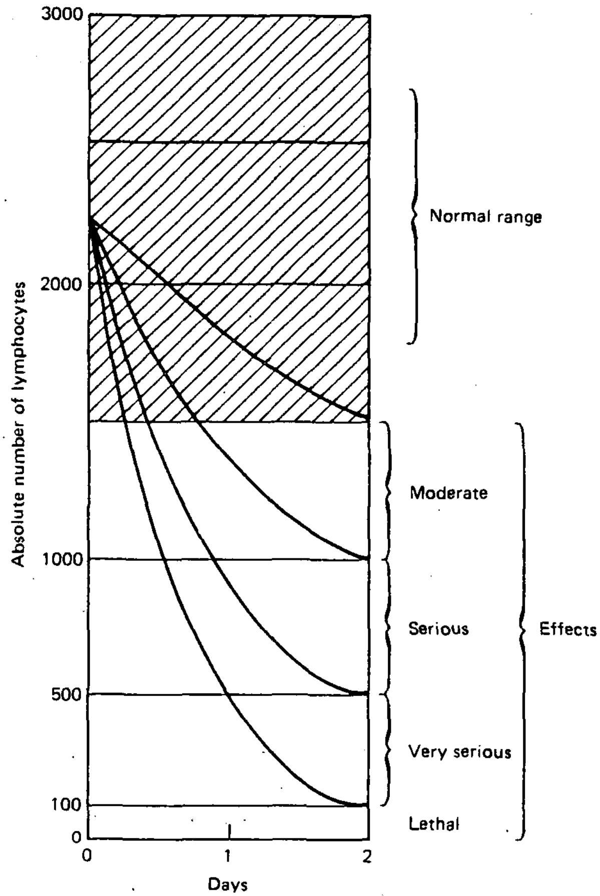


FIG. 3. Patterns of early lymphocyte response in relation to dose (from G.A. Andrews [5]).

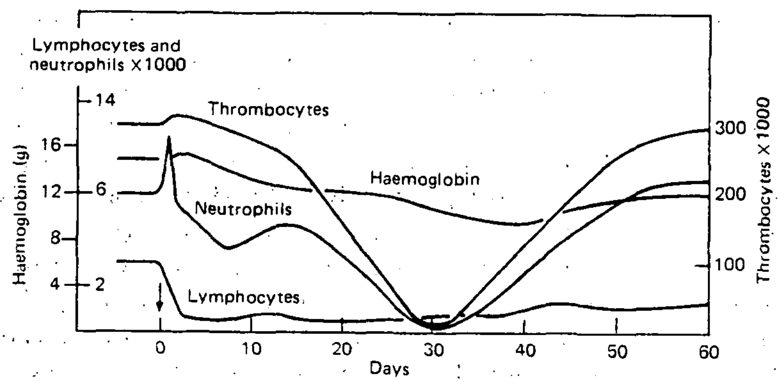


FIG. 4. Typical chart of blood values in fairly severe haematopoietic syndrome (from G.A. Andrews [5]).

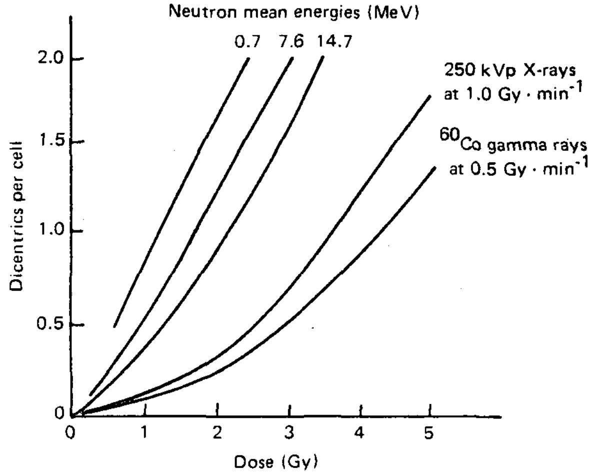


FIG. 5. Relationship between dicentric yield and acute exposure to several types of radiation (from D.C. Lloyd [14]).

In general, the changes in the peripheral blood counts following acute whole body exposure may vary from mild to very important. A decrease in lymphocytes occurs promptly, most of it taking place within 24 hours after exposure (Fig. 3). The level of this early lymphopenia is one of the best indicators of radiation dose and of severity of radiation injury. The level of neutrophil granulocytes shows a very early rise, usually limited to the first 48 hours or less; the degree of its elevation has not been correlated with the extent of injury (Fig. 5). (In the neurovascular syndrome, granulocytosis is very pronounced and persists until death.) In the dose range that produces the haematopoietic syndrome, after the early rise, granulocyte numbers fall to fairly low levels at about day 10, and there is then a transient, abortive rise at around day 15, perhaps due to mitoses of a genetically damaged population that cannot continue to reproduce. (The absence of an abortive rise is an unfavourable sign.) Then a steady fall in granulocyte count begins, with the nadir at about day 30 after exposure. A spontaneous recovery of granulocyte count beginning in the fifth week signifies good chances of patient recovery and survival. The platelets may show a rise in the first two or three days after exposure, then a gradually accelerating decrease with the nadir also reached at about day 30. During recovery the platelets are likely to bounce up well above normal levels.

(iii) *Bone marrow biopsy.* In cases of possible serious whole body exposure or prodromal symptoms and clinical signs of an expected radiation syndrome (sickness) bone marrow biopsy should be carried out to estimate the bone marrow dose and its distribution in space and time and for decision making in prognosis and treatment strategy.

Quantities and distribution of bone marrow effects are to be determined 24 hours after exposure (or at a time determined in respect to the reference curve of the laboratory) by cytological and cytogenetic tests.

Bone marrow samples should be taken from different localizations and repeated, taking into account the preliminary diagnosis.

(iv) *Biochemical.* Several attempts have been made to use biochemical indicators for radiation dose assessment but none of them has been introduced into routine practice [6].

2.2.2. Protracted whole body exposures

Although the probability of accidental protracted whole body exposure is low, cases have occurred, e.g. following the loss of a sealed source which subsequently was located unrecognized in the environment of the victims [7]. In such cases of prolonged irradiation, evaluation of the average absorbed dose and distribution, and the determination of subsequent treatment are complicated by:

— The difficulty of reconstructing the event.
— The absence of an initial blood picture and, hence, the change in the lymphocytes and the number of polynuclear neutrophils.
— The difficulty of interpreting the karyotype: the presence of several generations of irradiated lymphocytes makes it difficult or even impossible to use reference curves. Moreover, as in the case of the blood count, the results may be distorted by transfusions given after the exposure.

2.3. PARTIAL EXPOSURE

Partial exposure occurs when a sizeable portion of the body, mainly the head, chest and/or abdomen, is irradiated. The management of such cases in general is similar to that of whole body irradiation. However, prognostically the unirradiated parts of the haematopoietic system, through their capacity to repopulate the depleted irradiated bone marrow, have a profound beneficial effect on recovery.

In many accident situations irradiation of localized areas of the body could take place, such as by holding a source in a hand; when using old crystallography apparatus exposure of hands or forearms to the main beam can occur [8].

2.4. LOCALIZED EXPOSURE

Localized exposure occurs when a small area of the body, usually the extremities (hands, forearms), is irradiated from exposure to a narrow beam or by holding a radiation source, e.g. in the hand or in a hip pocket [9, 10]. Localized irradiation, however severe, does not lead to the occurrence of acute radiation syndrome.

2.4.1. Clinical symptoms

An initial sensation of heat, the onset of pains and paraesthesia and disturbances to tactile and heat sensitivity are the most commonly reported clinical symptoms. The initial sensation of heat depends not on the nature or energy of the incident radiation but on the dose and dose rate. The pain is almost constant and is perceived as a burning sensation throughout the irradiated part or a feeling of contusion or of tingling as in an electrical shock. It is very intense, constant and accompanied by paroxysms during the acute phase and is resistant to the major analgesics in serious cases.

2.4.2. Clinical signs

The scale of severity of the physical symptoms is the same as for common burns: erythema, oedema, blisters, ulceration, necrosis and sclerosis.

The thresholds at which these various lesions appear depend on the magnitude of the field. Following acutely delivered single doses for 3 cm diameter fields, the threshold doses are in the following ranges:

— erythema:	between 3 and 10 Gy
— dry epidermitis:	between 10 and 15 Gy
— exudative epidermitis:	between 15 and 25 Gy
— necrosis:	from 25 Gy.

These values increase as the size of the field decreases. They refer to single doses. Protracted or fractionated doses would have less effect than the same dose received acutely.

The clinical latency times for doses from 12 to 20 Gy are approximately three weeks for exudative epidermitis and from 6 to 18 months for necrosis.

2.4.2.1. Erythema

Erythema may be either:

— early and transitory (calling for careful examination), or
— secondary, immediately preceding exudative epithelitis, or
— late, after 6 to 18 months; at this stage erythema heralds increases in vascularization with oedema and pain syndrome.

2.4.2.2. *Oedema*

The topography of the early oedema gives indication of the volume irradiated, but does not enable its boundaries to be demarcated precisely. Usually the oedema is hot, hard, red, taut, shiny and accompanied by paraesthesia. The initial oedema appears between the first and the 21st day; the higher the dose the earlier its appearance. However, no clear correlation can be established between the dose and the date of appearance. The late oedema (several weeks, months or years after exposure) precedes and accompanies increases in vascularization. Its duration varies.

2.4.2.3. *Blisters*

These are characteristic lesions affecting the basal layer of the epidermis. Topographical and dimensional analysis of blisters is of interest for dosimetry. The periphery of the blister corresponds to a skin dose of between 15 and 25 Gy. The chronology of blister development depends on the dose received by the basal layer — the higher the dose, the shorter the time before the blisters appear: a lag time of 21 days corresponds to a surface dose of 12 to 20 Gy.

The blisters subside and heal, evolving then — as a function of the dose absorbed in depth — either towards cutaneous restoration or towards necrosis.

2.4.2.4. *Necrosis*

This occurs at doses above 25 Gy. It begins with ulceration, the ulcers formed being fairly deep with a yellowish base covered with a fibrinous exudate. Their appearance may change, becoming blackish when dried out (dry gangrene). The onset of necrosis occurs from several weeks to several months.

2.4.2.5. *Sclerosis*

Sclerosis occurs in muscular, tendon and aponeurotic tissues. The higher the dose, the earlier it sets in. As it is retractile, it presents functional problems, chiefly in the fingers and palms of the hands. It may lead to deformity, the hands becoming fixed in a clawlike position with the fingers hooked. In addition, annular sclerosis can develop on the phalanges causing oedema distal to the constriction.

CHAPTER 3

PROCEDURES TO FOLLOW IN THE EVENT OF AN EXTERNAL OVEREXPOSURE

3.1. SEQUENCE OF ACTIONS

The physician responsible for first aid must take as quickly as possible the following sequence of actions after the accident:

(i) evaluating the circumstances of the accident with the physicist and assessing the clinical status of the patient;
(ii) taking necessary therapeutic measures;
(iii) collecting biological samples;
(iv) evaluating dosimetric information supplied by physicists; and
(v) contacting a specialized institute.

3.2. CO-OPERATION OF PHYSICIANS AND PHYSICISTS FOR DOSE ASSESSMENT

The physician's role is to obtain information regarding the possible magnitude of the dose received by questioning in detail the accident victim and/or the authorities concerned. It is usually up to the physicist or team of physicists to collect the necessary information regarding physical dosimetry. It is the dialogue between physician, physicist and accident victim which enables the conditions of the exposure and the resulting doses to be evaluated as precisely as possible.

Combining the information obtained from the patient with the result of the dosimetry provides a means of interpreting as accurately as possible the results obtained from individual dosimetry. The main parameters such as the physical characteristics of the source and the geometry of the exposure and its duration may be well known in a given case, but the parameter of exposure time — of necessity subjective — is usually difficult to assess.

3.3. PHYSICAL DOSIMETRY

3.3.1. Evaluation of dosimeters

3.3.1.1. Personal dosimeters

The earliest dosimetric results are those obtained immediately from the personal dosimeters carried by the subject.

In assessing this information it should be remembered that:

— the exposure is rarely uniform;
— the dosimeter may have been in a place protected from the radiation field;
— the exposure may have been mixed, including for example a substantial neutron component.

The film most widely used consists of a photographic emulsion in a badge comprising several shields. It is used for determining the dose, the nature of the radiation (beta, X- or gamma radiation), the presence of thermal neutrons, the average energy in the case of electromagnetic radiation and the subject's orientation in relation to the source, i.e. whether turned towards it or away from it. If there is a risk of exposure to neutron radiation, this system must be supplemented by a dosimeter sensitive to neutrons. The persons exposed usually wear a chest film badge, but in some cases a wrist badge is also carried.

Thermoluminescence dosimeters are most often used for evaluating the dose (X- and gamma radiation) and could be used to determine the dose to the extremities (finger dosimeters).

Radiophotoluminescent glasses are used to determine the gamma dose. They are arranged in special housings between the film and the criticality badge.

The criticality dosimeter, which comprises various activation detectors, is attached to the badge containing the above-mentioned dosimeters and gives a first approximation of the neutron dose.

The criticality belt, which consists of photoluminescent glasses together with activation detectors arranged at regular intervals on the belt, is used chiefly for determining the orientation of the subject in relation to the radiation source and provides information to supplement the observations based on ^{32}P activation in superficial body features (hair, nails, teeth, etc.).

3.3.1.2. Area dosimeters

Area dosimeters are installed in any facility where there is a risk of accidental exposure. They are designed to enable the radiation field to be determined at the place where the contaminated person was irradiated.

3.3.2. Reconstruction of the accident

This may be done either experimentally or theoretically for the purpose of evaluating the dose distribution in the irradiated body.

3.3.3. The use of neutron activation for dose assessment

In addition to taking immediate readings of all the recoverable dosimeters, a number of further steps are necessary, especially where neutron or mixed exposure is concerned. In this case the first step is to measure as soon as possible the activity of ^{24}Na induced by neutron irradiation in the body of or in objects worn by the patient.

3.4. CLINICAL INVESTIGATIONS

The clinical observation of a person overexposed to radiation should begin as soon as possible since neurological and gastrointestinal symptoms or erythema of the skin give an early indication of high level exposure.

Clinical records made before the accident as part of the routine surveillance of personnel provide information which can prove helpful after the accident. The result of examinations carried out after the accident may then be compared with the results considered normal for the individual or alternatively with the range of values considered physiologically normal.

Clinical observation of an irradiated person does not differ from the procedures normally used in medical surveillance. The physician should follow the procedures he considers to be the most appropriate and which he normally employs. The clinical examination should be *as detailed as possible* with particular emphasis on:

(i) *Clinical history.*
(ii) *Repeated examination of the skin.* This is aimed at detecting signs of oedema or erythema. Note should be taken of the appearance of the affected tissue and length of time the erythema persists.
(iii) *Neurological examination.* The state of the patient in relation to consciousness, apathy and existence of fever should be checked. Abnormal responses relating to balance, co-ordination and the reflexes indicate exposure to a high CNS absorbed dose. It is also recommended that an EEG recording be made as early as possible. The changes in EEG can be used as a biological dosimeter.
(iv) *Cardiovascular examination.* The patient's blood pressure is measured and its evaluation recorded. An electrocardiogram may be taken to provide a record of the patient's post-accident condition.
(v) *Examination of the digestive tract.* The early onset and persistence of nausea, vomiting and diarrhoea indicate a high level of exposure.
(vi) *Ophthalmological examination* with slit lamp, which enables a baseline evaluation to be made of the crystalline lens and the anterior chamber. This examination must be carried out after whole body exposure or where exposure of the head is suspected and should be periodically (once a week) repeated.

The clinical examination will be of limited value unless it is repeated frequently, especially if it is suspected that the patient has been exposed to a high dose. It will be all the more useful if the chronology of all the symptoms observed has been carefully noted and if appropriate *reference photography* has been carried out. Photographs should be taken of all the regions likely to have been overexposed. Where a limb or extremity has been exposed, photographs should also be taken of the corresponding limb or extremity which has not been exposed. Photographs may need to be taken from several angles, particularly in the supposed position at the time of the accident. When the hands have been irradiated, for example, the palms and the backs of the hands should be photographed.

If there are no clinical symptoms, but it is at the same time certain that the patient has been irradiated, it may be that the average absorbed dose, although not inconsiderable, has a lower value than the threshold at which clinical symptoms appear. This should be confirmed, however, by blood and urine samples taken for the purpose of conducting biochemical, cytological and chromosome tests. The results of these will allow the level of absorbed dose to be estimated more reliably.

3.5. BIOLOGICAL INVESTIGATIONS

3.5.1. Immediate action

It is important to carry out an immediate blood test (total and differential blood count) and to take (and refrigerate) a urine sample. Provided these are carried out before the overexposure has given rise to any changes, these samples provide extremely useful information on the values normal for the patient. The latter values are the best basis for comparison with those obtained subsequently.

3.5.2. Blood samples

The blood is used for three different types of test:

— haematological: total and differential blood count, including platelets; haematocrit, haemostasis, etc.;
— biochemical: electrolyte balance (at least);
— chromosome analyses: lymphocyte culture for the investigation and counting of anomalies (dicentrics, rings, etc.).

It is important to repeat the total blood and leucocyte counts a number of times during the first day. By conducting a test every three hours it may be possible to detect an 'early leucocyte peak' occurring during the first 24 to 36 hours. If these tests cannot be evaluated on the spot, it is still helpful to repeat the sample on ethylenediaminetetraacetic acid (EDTA, 2 to 5 mL) a second and third time within the first 24 hours. Subsequently, samples for the haematological tests should be taken every day at the same time.

3.5.3. Urine samples

The urine samples taken as soon as possible after the accident are of particular value. It is important:

— to note the daily urine volume;
— to take a sample from the first urination and to note its time precisely;
— not to mix the urine samples with those from other urinations during the same day;
— to label the samples carefully.

The samples can be preserved by refrigeration, if possible in the freezing compartment. No other means of preservation are necessary.

3.5.4. Further tests

Other tests may sometimes be necessary. These include:

— Thermal methods: thermometry
thermography
— Radioisotope techniques: vascular scintigraphy
— Physicochemical techniques: measurement of body fluids
capillaroscopy
measurement of circulation velocity.

Two types of tests should be given priority and should be performed whenever possible: telethermography and vascular scintigraphy using $^{99}Tc^{m}$. These are particularly indicated where lesions are localized in an extremity, e.g. lesions of the hand.

3.6. GENERAL PRINCIPLES REGARDING TREATMENT

The modes of treatment depend on whether the exposure has been whole body or localized.

3.6.1. Whole body exposure

The management and treatment of individuals who are exposed to considerable whole body exposure (WBE) are closely similar to management of patients with bone marrow aplasias and those of immunosuppressed patients. Measures must be taken to avoid any superinfection. Careful examination of all common foci of infection (ears, teeth and throat) and the gastrointestinal tract should be carried out including bacteriological tests, and steps be taken to keep the patient in a good state of hygiene.

These should be carried out in the early period before the patient progresses into a critical phase. Fungal infections generally lead to serious consequences. It is essential that the individual be isolated in special facilities (see Section 5.4.1). As stated earlier, a major consequence of considerable WBE is bone marrow aplasia which becomes severe around four to five weeks after exposure. The onset of agranulocytosis and thrombocytopenia occurs early and it is preferable to correct these with suitably matched transfusions of granulocytes and blood platelets and thus compensate their fall and prevent infectious haemorraghic episodes.

The actual treatment may be vicarious or compensatory. The experience with transplantation of allogenous bone marrow after the Chernobyl accident gives rise to two important conclusions for the future in respect of bone marrow transplant (BMT) treatment:

- in radiation accidents the proportion of patients in whom transplantation of allogenous bone marrow is absolutely indicated and for whom this treatment will obviously be beneficial is very small;
- with spontaneous recovery of bone marrow caused by overall gamma radiation doses of the order of 6–8 Gy, a transplant may 'take' but this 'taking' will frequently have a negative effect in therapeutic terms and even endanger life as a result of the high risk of secondary disease developing.

The latter conclusion is essentially a new one since it was assumed earlier that the transplantation of allogenous bone marrow does not give rise to negative effects in the event of insufficient radiation exposure of the recipient in the borderline zone of radiation doses.

Therefore, in general, human leukocyte antigen (HLA) identical bone marrow transplantation seems indicated only in cases of uniform whole body irradiation at dose levels of 9–11 Gy or more.

For this reason the compensatory treatment is now usually preferred. This method involves transfusing concentrates of elements of the deficient cell lines. In particular, it is necessary to control anaemia, agranulocytopenia and thrombocytopenia. In the case of serious and persistent aplasia, the necessity of carrying out repeated and large transfusions of erythrocyte, leucocyte and platelet deposits in order to maintain these cells at acceptable levels, while at the same time limiting the production of antibodies which would prohibit any marrow transplant, can present considerable practical problems.

The treatment of protracted exposure is, broadly speaking, the same as that of acute exposure. It is dictated by the length of the critical period and the extent of medullar depletion manifested by anaemia, granulocytopenia, thrombocytopenia together with infectious complications which may present a particular threat (e.g. endogenous infection by fungus and septicaemia caused by intestinal bacteria).

3.6.2. Partial body exposure

Such exposure could lead to a clinical picture similar to the whole body irradiation syndrome. The clinical management is the same as ARS. However as parts of the haematopoietic system escaped irradiation the haematopoietic syndrome is milder with good prognostic recovery expectation.

The signs and symptoms of partial body irradiation as observed in some of the Chernobyl cases of external irradiation are summarized in Appendix 4.

3.6.3. Localized exposure, mainly of extremities

The aim of the treatment of localized injury is to promote rehabilitation and to prevent further traumatization, and therefore no biopsy in general is indicated. The injured area should be locally isolated. During the first week after injury a guarded prognosis should be given.

Localized exposure leads to pathological effects in different tissues, depending on the quality of the irradiation in question and the distribution of the dose in depth and the size of the irradiated surface.

(i) *Superficial irradiation*

Recovery is normally spontaneous. To limit trophic sequelae (i.e. sclerosis), various treatments are proposed:

- Daily changing of the dressing with:
  - bathing with antiseptic solutions,
  - active movement of the joints.
- Possible maintenance of the exposed part of the limb in a local isolator.

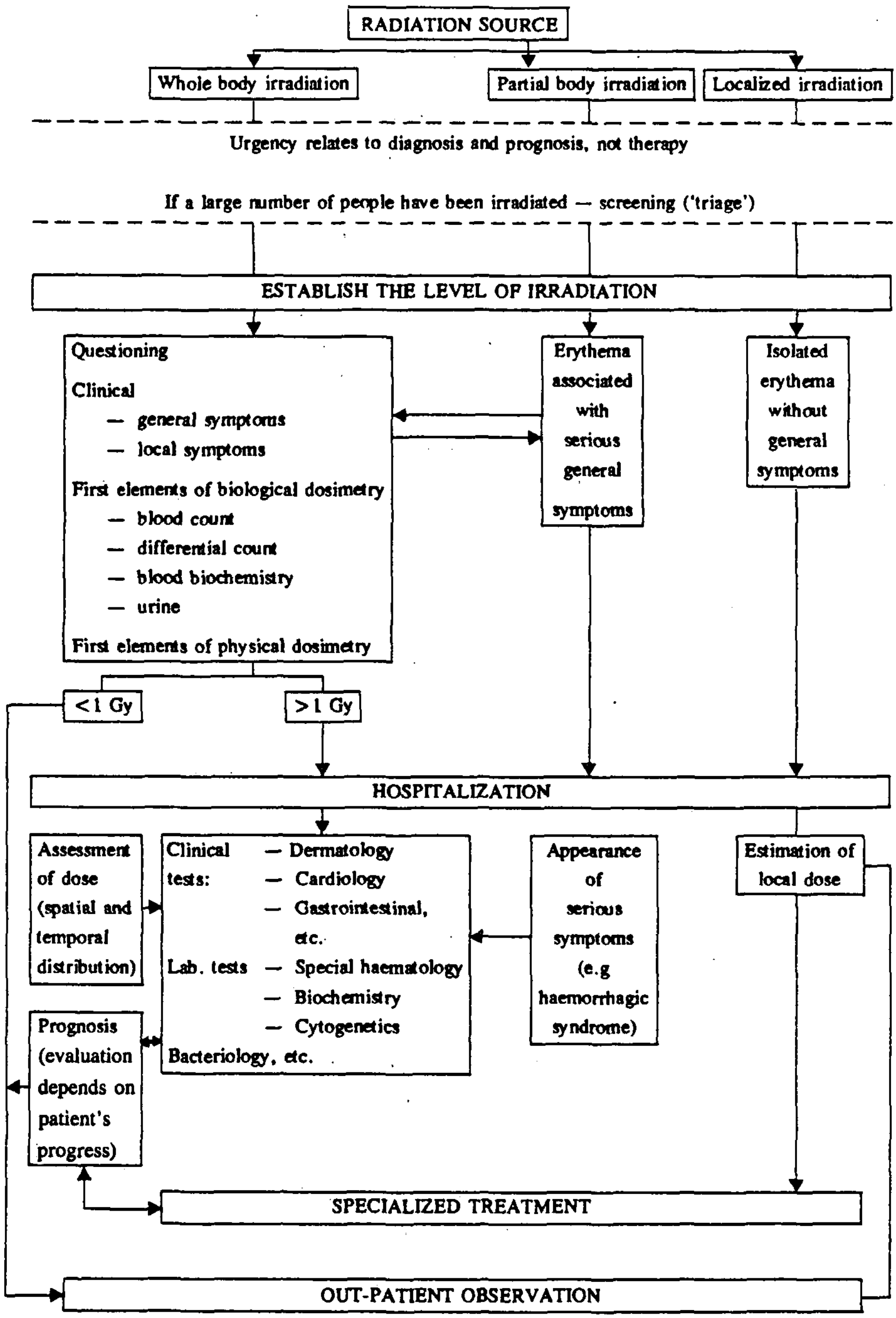
(ii) *Deep irradiation*

Deep irradiation may require surgical intervention. The excision must be complete (i.e. amputation) for distal necrotic parts. Where excision is incomplete the next step is immediately to cover the tissue with a myocutaneous flap where the excision was made.

This excision is made taking account of isodose curves and the appearance of the resection limit during surgery. All the necrotic tissue must be eliminated. A flap with an attached pedicle or one taken from a remote part of the body is used to cover the tissue exposed after surgery. Surgical intervention should be carried out as soon as possible in order to prevent the development of tolerance to the major analgesics and persistent phantom pains, although the decision for surgery is usually a difficult one as it is not easy to recognize deep lesions early in the absence of apparent clinical signs.

The treatment of the pain caused by irradiation presents serious problems. The pain is constant, intense in most cases and accompanied by paroxysms, and is

TABLE IV. MAIN STEPS TO BE TAKEN IN RESPONSE TO ACCIDENTAL OVEREXPOSURE TO A SOURCE EXTERNAL TO THE BODY



particularly intolerable when the patient has deep lesions due to high energy radiation. The pain is greatest when vascularity increases: it lasts several days and develops in surges lasting several hours per day. It is particularly resistant to normal treatment, even medication based on opiates. This pain may, by virtue of its intensity and persistence, become the major focus of treatment.

3.7. CONCLUSIONS

Table IV gives a summary of the main steps to be taken in response to an accidental overexposure to a source external to the body. This table provides a highly simplified outline of the most useful action which can be taken as various dosimetric, clinical and biological data are progressively obtained. The decision level of 1 Gy is not to be interpreted rigidly. It serves as a guide for the level of dose at which certain treatment decisions should be made.

The list is not exhaustive. A number of stages should be regarded as indispensable:

- — the immediate taking of blood and urine samples;
- — approximate activation measurements using conventional radiation protection instruments;
- — the collection, within as short a time as possible, of the results of personnel or group dosimetry;
- — the preservation of samples, especially from hair, nails, etc. for the measurement of beta radioactivity;
- — complete and repeated clinical examination.

Finally, it should be repeated that in most cases accidental overexposure is not regarded as an emergency in the sense of requiring immediate treatment but rather in the sense of requiring immediate dosimetric action or, at any rate, action aimed at gathering the data or taking the samples necessary for dosimetry. This is, therefore, the main focus of immediate efforts in the event of an accidental overexposure.

CHAPTER 4

RADIOACTIVE CONTAMINATION

4.1. GENERAL CONSIDERATIONS

The incorporation of radionuclides could result in stochastic and non-stochastic effects. The stochastic risks associated with the incorporation of radionuclides in different organs or the irradiation of different organs are given in Table V based on the ICRP Recommendations (ICRP No. 26) [11]. The stochastic risks include severe genetic effects over the first two generations. In addition, the stochastic effects for which the risk estimates are given are for fatal cancers. These risk factors are overall average population values. The ICRP, in order to prevent non-stochastic effects on organs following intakes of radionuclides which if based on the limit of effective dose alone would be acceptable, introduced a limit of 500 mSv for organ doses to prevent non-stochastic effects. The overall risk factor resulting from the irradiation of the whole body is 1.65×10^{-2} Sv^{-1}.

TABLE V. RISK FACTORS BASED ON ICRP 26[a]

Tissue	Risk factors Risk (Sv^{-1})
Gonads	4.0×10^{-3}
Breast	2.5×10^{-3}
Red bone marrow	2.0×10^{-3}
Lung	2.0×10^{-3}
Thyroid	0.5×10^{-3}
Bone surfaces	0.5×10^{-3}
Remainder	5.0×10^{-3}

(from ICRP Publication 26, 1977)

[a] The risk factor for hereditary effects can be evaluated as 4×10^{-3} Sv^{-1} in the first two generations. The total risk of hereditary damage in all subsequent generations is considered to be about twice that which is expressed in the first two generations.

The biological consequences of contamination depend on several factors, among which are:

— The portal of entry;
— The organ(s) of deposition;
— The nature of the emission from the contaminating radionuclide;
— The effective half-life of the contaminating radionuclide;
— The physicochemical nature of the contaminant.

The effective half-life (a combination of the radioactive half-life and the biological half-life) is something often forgotten in initial evaluations; it is significant, for example, that the effective half-life of tritium is about ten days whereas its radioactive half-life is twelve years.

4.1.1. Biological factors

Internal contamination involves four successive stages:

(a) *Intake.* Possible ways of intake are via the respiratory system, wounds, the gastrointestinal tract or the intact skin.
(b) *Uptake.* Absorption of the radioactive material into the extracellular fluids, such as blood or lymph.
(c) *Deposition.* The incorporation of radioactive material into tissues and organs.
(d) *Elimination.* The removal of radioactive material from tissues, organs or the whole body.

Treatment could theoretically be designed to operate at any of these stages, but, in fact, preventing the uptake to the extracellular fluids by fixation of the radionuclide at the site of entry or trapping it in the blood allowing rerouting towards a natural excretory mechanism constitute the two current methods of treatment. Treatment is aimed at stage (a) or stage (b). Action at stage (c), which would prevent deposition in a target organ, is possible in the specific case of the thyroid. Action at stage (d) is generally ineffective except for radionuclides like tritium which can be flushed from the body.

4.1.2. Influence of physical and chemical factors on biological behaviour

As a rough approximation, radionuclides can be classified on the basis of their behaviour in biological material into two categories: *transferable* and *non-transferable*. It should be noted that this classification is highly schematic and that the division between them is more apparent than real.

The radionuclides described as *transferable* are soluble in biological material and able to diffuse throughout the organism; the entire deposit may pass rapidly through the metabolic pathways, leading to deposition in the target organ. They are usually present in the organism in the physiological form — either with a stable isotope (for example, iodine) or a chemical analogue (for example, caesium–potassium or strontium–calcium). Their biological behaviour, especially their deposition in a critical organ, depends on the metabolism of the corresponding physiological analogue. Thus, iodine accumulates in the thyroid and caesium in the muscles, while strontium and radium follow calcium in bone hydroxyapatite. Schematically speaking, the materials in question are mineral cations of valences I or II; their ability to spread through the organism is a function of their dissociated *ionic form* at the pH of the biological material. Something to be noted is that they are capable, accordingly, of passing through the digestive tract.

Among such elements there are two large families of cations that have two different metabolisms: the *alkalis* (sodium, caesium, potassium, etc.), which spread through the whole organism, and the *alkaline earths* (calcium, strontium, barium), which accumulate in the bone structure. Caesium illustrates the importance of a knowledge of the metabolism if one is to apply effective treatment. It goes through a quantitatively significant intestinal cycle; secreted in the lumen of the intestine, it is almost completely reabsorbed unless the cycle is interrupted by a medicament present in the intestine which selectively complexes caesium (such as Prussian blue). Irrespective of what occurs at the nuclide's route of entry, such a drug reduces the caesium concentration in the blood and thereby speeds up the elimination of the radionuclide from the cells as an indirect effect.

Apart from these two large families, there are radionuclides exhibiting a specific type of behaviour, for example, carbon, iodine, tritium and the noble gases. Two radionuclides may give rise to special problems, namely carbon-14 and tritium. There are as many problems as there are labelled molecules; these radionuclides share the fate of the molecule or the fraction containing it if degradation occurs. In the case of carbon-14, for example, degradation generally yields $^{14}CO_2$ which must then be sought in exhaled air; otherwise only measurements of the excreta, coupled with an exact knowledge of the molecule's metabolism, can lead to an evaluation of the body burden.

The radionuclides described as *non-transferable* do not obey the criteria listed above, for either of the following two reasons:

(i) They are *insoluble* at all pH levels, like certain metals or oxides calcined at very high temperature which, for practical purposes, do not diffuse through the body at all. Actually, there is always some local diffusion, however small.

(ii) They are soluble only at acidic pH levels and their salts are hydrolysed as the pH rises, producing hydroxides which are polymerized on the spot. However, at the moment of contamination and during the time required for equilibrium to be established between the initial monomeric form of the contaminant and the final polymerized form, there may be absorption of a small fraction of the quantity deposited. From then on, absorption depends on the degradation of the polymers by the organism and is in any case a very slow process. For such elements the important target organs and tissues are, in addition to the point of entry, the liver and the bone surface (subendosteal and subperiosteal spaces). An example is plutonium-239 nitrate.

These elements are usually mineral cations with a valence higher than II. The rare earths, plutonium and the transplutonics are the most important examples. Their insolubility in the living organism disappears as they are present in complexed form and the complex is stable and soluble. Their biological fate is then that of the complex (and hence that of the chelating agent), rather than that of the element. This property is highly significant from the point of view of treatment; the elements are chelatable with the calcium salt of DTPA (diethylenetriaminepentaacetic acid) and can therefore be redirected towards elimination by the kidneys during their passage in the blood, thereby avoiding prolonged deposition in the liver or bone structure.

It should also be borne in mind that *solubility* is relative and always in direct dependence on the medium. The medium varies according to the organ concerned, both in terms of pH and of redox potential. For example, an element inhaled in an insoluble form may be made *absorbable* in the digestive system if it dissolves in the stomach acid, with dissociation of the salt. Conversely, a soluble element may be made completely insoluble in the digestive system by alkalinization in the duodenum, with the formation of insoluble hydroxides.

4.1.3. Availability of contamination data and decision on treatment

A major problem in the early management of persons contaminated with radionuclides is that the extent and magnitude of internal contamination is unknown. In complex contamination situations, particularly those involving alpha emitters, dosimetric evaluations may have to be delayed several days while the sequential samples of excreta or measurements of low energy photons of alpha emitters in the chest are being done. Conversely, treatment procedures are most effective if initiated soon after the contamination has occurred. As a result, the critical initial treatment decisions may have to be based on a knowledge of human physiology, the pharmacology and metabolism of the particular chemical compound, and whatever information regarding the exposure potential is available at the time. This judgement is best made after a detailed review of the exposure incident and with foreknowledge of the available treatment regimens and procedures that were developed in the preplanned emergency response for management of such accidents.

The earliest information after the exposure will consist perhaps of some information on the accident, probably identification of the major radioisotopes by history or early spectrometric identification, a few radiological measurements (contamination surveys, air concentrations, etc.), and no clinical symptoms or signs other than possible trauma. The probably complete absence of immediate clinical features places the physician at a great disadvantage in trying to determine the need for treatment. An initial survey for contaminating radionuclides is the first measurement required to indicate the probable magnitude of the patient's involvement in the accident. In accidents involving airborne materials, nasal wipes should be taken before the patient showers. These may serve as an index of possible inhalation exposure, although negative values do not eliminate the possibility of exposure. The physician should not delay making his initial decisions to start first aid procedures while waiting for better dose estimates. His decision will be based on rough exposure estimates, past experiences, clinical appraisal, and medical judgement.

After immediate first aid treatments, further diagnostic studies of value will be whole body counting for gamma emitters, if available, and urine, faecal, and blood samples for radioanalysis. With these initial results it may be possible to arrive at a somewhat better estimate of the exposure. This preliminary appraisal may be adequate to suggest additional treatment methods, such as chelating agents if they have not already been employed.

After initial treatment measures, a time interval is generally available for obtaining detailed physical assessments of external and internal contamination before embarking on additional therapeutic procedures. Advance emergency planning will reduce the time required for dosimetric evaluation because the plan should list available sources for appropriate calibration, spectrometric measurement instrumentation, and dosimetry and analytical chemistry services. This is also the time when medical and health physics consultants familiar with the management of radiological accidents can be brought in for consultation and advice.

The bases for treating the patient are common to all medical practice. The physician must judge the ultimate benefit versus the potential harm of the procedures he performs. This decision becomes more difficult when the potential hazard is a long delayed health impairment, the possible occurrence of which can be estimated only by applying the probabilities of developing subsequent biological effects based on an assessment of absorbed dose. The possibility of immediate or short term, as well as long term, health risks from the treatment must be balanced against this hazard. The judgement of the physician can lead to a range of actions, no treatment, débridement of contaminated tissues, non-specific supportive treatment, or a variety of medications, including long term drug therapy.

If it is assumed that the quantity of radionuclide entering the blood is proportional to the diffusible quantity present at any moment at the point of entry, the effectiveness of local treatment decreases exponentially with time. Hence, it is important to treat contamination at the point of intake especially when the radionuclides concerned are those for which no effective therapy is available once they have been absorbed. Since deposition in the target organs begins as soon as the radionuclide is present in the blood, the amount deposited increases inevitably with time. As a result, even in the case of radionuclides accessible to treatment at blood level or by preliminary blocking of deposition in the target organ (for example by stable iodine), time remains important.

The methods of treatment will vary depending on the route of entry of the contaminant. Consideration must be given to the different possibilities: contamination of the skin, contaminated wounds and internal contamination.

4.2. CONTAMINATION OF INTACT SKIN

4.2.1. Objectives of treatment

The objectives of skin decontamination are to remove as much of the radionuclide as practicable in order to reduce the surface dose rate and to prevent activity from entering the body. Careful skin decontamination can also enhance the accuracy of whole body counting for estimation of internal body burdens. An overaggressive skin decontamination effort must be avoided since it may injure the natural barriers in the skin and so increase percutaneous absorption.

4.2.2. Physical and biological principles

Many cases of skin contamination with radioactive materials will be decontaminated by non-medical personnel at or near the accident scene. When initial cleansing methods are not effective, the patient should be referred to a physician. The physician's decisions on decontamination procedures should be based on an understanding of the special physical and biological principles involved. Success in achieving the objectives stated above requires thoughtful appraisal of the level of residual contamination, rate of successful decontamination, and condition of the skin. These factors change continuously as the cleansing procedures proceed.

The full thickness of the skin is about 2 mm. Of this, the epidermis has a depth of about 0.1 mm in most parts of the body. On the palms and palmar surfaces of the fingers it may reach 0.8 mm and on the sole and toes of the foot, 1.4 mm, due to the thickened horny layer or stratum corneum. For the estimation of dose to the skin, the relevant tissue is the basal cell layer which is located at an average depth of about 0.07 mm except on the palms and soles as noted above. On the face the depth of the basal cell layer is somewhat less than in the rest of the skin. The small blood vessels of the dermis can be injured when energetic beta, X- or gamma rays impinge on the skin in high doses — several Gy or more.

The possibility of biological effects from skin contamination varies with different types of radiation and their energies. With alpha radiation, the stratum corneum effectively shields the basal layer. For example, plutonium alpha rays with a range of only about 0.04 mm in soft tissue do not reach the basal cell layer of the epidermis. Contamination with alpha emitters is of concern, however, because of the possibility of percutaneous absorption and possible irradiation by beta or gamma emission from daughter products. Loose contamination with alpha emitters on skin may also result in an ingestion and inhalation hazard.

One millimetre of tissue will reduce most beta radiations by a factor of two or more so the dose from surface beta contamination to the subcutaneous tissue is much less than that to the epidermis. Beta radiation with an energy less than 50 keV does not penetrate the outer layer of the skin and even at energies up to 200–300 keV, much attenuation occurs.[1]

Low energy gamma radiation (10–15 keV) has an effect on the skin similar to beta particles. Gamma radiation of 50-300 keV from a source on the skin is capable of producing biological effects in the deeper layers of the skin and in the subcutaneous tissue.

Percutaneous absorption is of concern with all types of radionuclides and, when it occurs, most chemicals pass through the epidermis rather than its sweat glands and hair follicles. The most important barrier to penetration is the horny layer, which has been described as a 'hydratable fibre mat' rather than a set of membranes. When contamination reaches the bottom of this layer it diffuses rapidly through the rest of the epidermis and into the capillaries of the dermal papillae. The lower layers of the stratum corneum possess a sponge-like capacity to fill and empty. This process may explain why alpha contamination can sometimes 'reappear' after decontamination. The level of residual alpha contamination on the surface of the skin may therefore increase 24–48 hours after initial satisfactory decontamination.

Contaminants may be held to the surface of the skin by electrostatic forces or surface tension. Also, chemical compounds may be formed between the contaminant and skin protein. Mechanical entrapment in the hexagonal plates of the horny layer may occur with material of small particle size. Although contamination may sometimes be extremely difficult to remove by physical or chemical means, the skin will cleanse itself, the horny layer being shed and renewed on a 12–15 day cycle. The turnover rate differs considerably in various areas of skin. Forehead skin appears to turn over faster while on the back of the hand it is slower than for other parts of the body. Any damaging influence such as ultraviolet light, stripping with sticky tapes, or scrubbing reduces the renewal time. In general, most contamination is limited to the upper part of the horny layer and will therefore usually be shed in just a few days.

[1] The dose rate at the basal layer from 37 kBq of ^{14}C, average beta energy of 50 keV, uniformly distributed over 1 cm^2, is 17 $mGy \cdot h^{-1}$, while for the same amount of ^{32}P, average beta energy of 695 keV, it is 64 $mGy \cdot h^{-1}$. The surface activity of beta emitters required to produce acute skin damage is thus highly variable. An acute irradiation of the skin at high dose rates (>0.3 $Gy \cdot min^{-1}$) with 200 kVp X-rays produces erythema at about 6 Gy and moist desquamation at about 20 Gy.

4.2.3. Evaluation of contamination

After an accident has occurred, the initial radiation survey of the skin should be made with both a beta–gamma and an alpha monitoring instrument.

Skin contamination with beta emitters of even moderate energy can sometimes be missed by a person inexperienced in the proper use of radiation monitoring equipment such as a field survey instrument. Survey starts with an open shield so that both beta and gamma will be detected. Beta emitters are capable of producing beta burns if heavy contamination occurs and if the contaminated person, unaware of its presence, allows it to remain on the skin or surgical glove too long.

Another method for determining the distribution of radioactive material on skin is the high speed alpha autoradiography technique in which the contaminated area is placed against a zinc sulphide scintillation screen and Polaroid film for a few minutes exposure time. The time of exposure needed for best visualization depends on the level of contamination and may take some experimentation.

4.2.4. Decision levels

The physician will want to know when the decontamination effort can be safely stopped and also when it is unwise to continue. Citing an absolute numerical level would require so many assumptions as to be misleading. Typical questions that arise are: Is the contamination confined only to the superficial parts of the horny layer? Has it penetrated uniformly throughout the layer, or has it already penetrated through to the basal layer? If the contamination is on or near the surface, it will probably be sloughed in two or three days. If it is uniformly distributed throughout the horny layer, the rate of sloughing will be similar to the turnover time of the horny layer itself, about six to seven per cent per day. The rate of absorption depends on the physical and chemical characteristics of the contaminant. For these reasons, it is not realistic to set down arbitrary radiation levels that would indicate whether or not to pursue additional decontamination.

It is always wise to remove as much as possible without seriously irritating the skin. Most workers will be apprehensive if they can see a positive instrument reading. Saying that such a count rate is inconsequential and decontamination is not needed will not satisfy most patients. After a reasonable effort has been made, it is much easier to convince the patient that further efforts are not necessary and might be counterproductive.

4.2.5. Principles of skin decontamination

Spreading of contamination through the organism should be prevented if at all possible. The rule is to avoid abrasion of the skin and the use of products that could facilitate the passage of material through the skin. The radionuclide should thus be removed by washing, by solubilization or by the application of strippable material to the skin. There are special indications for each type of decontamination. Generally speaking, preliminary decontamination of the skin is carried out on the spot; verification of removal and, if necessary, more elaborate treatment to remove residual or stubborn contamination are carried out at the medical facility at the place of work.

Soaps and detergents emulsify and dissolve contamination and are frequently all that are needed for decontamination of skin. Gentle brushing or the use of an abrasive soap or abrasive granules dislodges some contamination physically held by the skin protein or removes a portion of the horny layer of the skin. Addition of a chelating agent helps by binding the contaminant in a complex as it is freed from the skin. Sodium hypochlorite (household bleach) has been found useful for several plutonium compounds. Titanium dioxide has an abrasive action and should not be used near the face or sensitive areas.

If long lived alpha emitters are the contaminants, contaminated hair should be removed by clippers or an electric razor. The eyebrows should be clipped only if further removal of contamination is deemed essential. The eyebrows regrow slowly and may take six to eight months for return to normal. Hair clippings should be collected in a plastic pouch and surveyed for contamination. When hairy areas have been contaminated with beta-gamma emitters and absorption through nicks, cuts, or scrapes is of less concern, shaving with safety razor and soap will often remove most of the contamination.

When using any of the above agents or techniques, one should always begin with the least irritating and proceed to stronger or more abrasive techniques only if absolutely necessary. It should be recognized in this sequence that the stronger agents are used after a certain amount of skin irritation has already occurred. It is a common mistake to underestimate the potential for skin irritation until too late. Particular care should be taken on the more sensitive and thin skin areas, such as an area that has been skin grafted or used as a donor site. Progress should be checked by frequent resurveys with the radiation detection instrument. If long lived alpha contamination is left on the skin to be shed by the normal renewal process, it should be covered with a dressing. Rubber gloves are particularly useful for this when the hands are contaminated. Each day when the covering is removed, both it and the skin should be monitored for the amount shed and the residual contamination which remains.

While chemical techniques are seldom necessary, they can be used. Potassium permanganate, a powerful oxidizing agent, and sodium acid sulphite, which removes the permanganate stain, will remove a portion of the horny layer. In especially stubborn cases where the contamination seems to be localized in one or two tiny spots in the thick horny layer, such as the palms of the hands, abrasive methods can be used to sand off the spot. Sticky tapes have been used successfully, but they tend to remove the horny layer rapidly and, if not used carefully, can lead to increased percutaneous absorption.

4.3. DAMAGED SKIN CONTAMINATION

In the case of contaminated wounds or burns an initial assessment of the following should be made:

— The *severity* of the injury. Emergency treatment of the injury takes precedence over treatment of the contamination.
— The *degree* of the contamination.

After emergency first aid, if required, has been given to control haemorrhage and treat shock, the crucial first step in decontaminating wounds is to find the contamination. With energetic beta emitters and most gamma emitters this is not a major problem but with weak beta and alpha emitters the detection can be difficult. In most cases the removal of contamination, once found, presents no special surgical challenge but does require considerable patience and perseverance.

4.3.1. Uptake and deposition of contaminant

The uptake from a wound to the general circulation or the deposition in regional lymph nodes is of major concern. The physicochemical characteristics, e.g. solubility, pH, tissue reactivity, and particle size, will determine the speed of movement from the wound. If the contaminant is in a highly acidic or caustic state, the tissue proteins may coagulate and reduce dispersion into the tissue fluids. Some radionuclide compounds will gradually change their solubility after prolonged contact with body fluids. Plutonium chloride or nitrate is soluble in an acid pH, but in the slightly alkaline biologic milieu it transforms into hydroxides that polymerize into non-transferable aggregates. Plutonium oxides, especially those that have been high fired, are considered rigorously insoluble, yet after prolonged exposure to body fluids they become partially soluble (transferable). In general, long lived radionuclides such as ^{239}Pu and ^{90}Sr are of greatest concern because they continue to irradiate the surrounding cells intensely at the wound site or in the internal organs, if translocated.

Dog experiments indicate that insoluble PuO_2 in the form of small particles can be rapidly translocated. Thus, prevention of movement of even an insoluble compound requires prompt action.

4.3.2. Classification of skin injuries

Abrasions. A contaminated abrasion presents considerable potential for absorption since the surface is often raw and bleeding, and the epidermal barrier is no longer intact. Usually such surfaces can be cleaned with a detergent and, if necessary, a topical anaesthetic, such as 4% lidocaine, can be used to allow more vigorous cleansing. After a reasonable effort, there is no need to attempt to remove all contamination since the residue that remains on the surface will probably be incorporated in the scab. When the scab sloughs it should be saved for measurement of radioactivity and proper disposal.

Punctures. Punctures may result from contaminated metal or glass slivers, small tools, or accidentally by hypodermic needles broken during injection. In explosions a small missile may be driven through the skin and may leave only a small entry wound. Its exact position may be difficult to locate and thus require considerable surgical excision of the wound.

Lacerations. A simple clean laceration made superficially by a contaminated sharp object is probably the least difficult type of wound in which contamination has to be detected and then decontaminated. Often much of the contamination is deposited on the lips of the wound. When lacerations are ragged and deep, contamination may be deposited in fascial planes with subsequent migration that makes difficult the detection of the contamination and subsequent decontamination into a blood vessel or major lymph channel.

Burns. Contaminated burns present considerable potential for absorption since the surface is often raw and bleeding and the epidermal barrier is no longer intact. Primary attention should be given to the treatment of the burn and, if appropriate, the contamination could be treated as mentioned in abrasions, punctures or lacerations depending on the depth of the contaminant and the extent of the burn. However, caution should be exercised to avoid vigorous rubbing or cleansing. In any case most of the insoluble contaminant will be shed with the scab. Chernobyl experience has shown that primary beta burns can occur in accidental conditions. These cases should be treated in the usual way as contaminated thermal and chemical burns. Perhaps more frequent dressings should be used during the first few days to allow the contaminant to be shed with scabs and dressings themselves.

4.3.3. Precautions

Large amounts of beta-gamma emitting contaminants may present a radiation hazard to physicians, nurses and other attendants. The potential exposure situation can always be evaluated rapidly with portable beta-gamma survey instruments. Improvised shielding may be necessary if a special shielded decontamination facility is not available. In order to estimate the skin exposure on the hands of the surgeon, thermoluminescent dosimeters can be taped at a location on the palmar side of the hand that will not interfere with tactile sensation or grip. If the contaminant is a weak beta emitter such as ^{3}H or ^{14}C, double gloves should provide sufficient protection.

4.3.4. Treatment of wounds and surgical considerations

When it is determined that the patient has radioactive material in a wound, efforts to clean the wound should begin in a manner similar to cleaning a dirt laden wound or removing a foreign body. Irrigation of the wound with sterile water or saline, free bleeding, and occluding venous return with a tourniquet have been advocated for immediate action.

If the patient's general condition requires first aid procedures or medical therapy for serious complications, they shall take precedence over the decontamination effort. Another early consideration of importance is determination of the possible advantages of administering blocking agents and/or the use of isotopic dilution. If the contaminant is plutonium or one of the other long lived alpha emitters for which DTPA is an effective chelating agent, treatment should be started promptly.

After administering suitable drug therapy as indicated, further decontamination and surgery may be considered. Thorough irrigation with normal saline will often flush out much of the contaminating material. A pulsating water jet lavage has been used with some additional success. The wound frequently needs to be enlarged for better irrigation and it may be necessary to excise a block of tissue in order to remove most of the contamination. Enlarging the wound with an incision on each side and under the wound and then lifting out the block of tissue is often effective. If the initial laceration was jagged or there was much tissue destruction, this technique may provide a wound that is easier to close. Use of a skin biopsy punch is a convenient way to excise small puncture wounds. Serious attention should be given to the cosmetic outcome of surgical decontamination, specially if the face is involved. As surgical instruments become contaminated, as determined by frequent monitoring, they should be removed from the surgical field and replaced with fresh, sterile tools in order to prevent extension of contamination. The difficulties involved in detecting the contaminant, while preventing its spread, will mean that the surgical procedure may last much longer than normally expected. Many more instruments and sponges than usual will be required. Even though pulmonary contamination has occurred, inhalation anaesthesia and respiratory assistance apparatus should be used without regard to possible risks of contaminating the equipment. Only those portions of the equipment directly in contact with the mucous membrane of the respiratory tract are exposed to a significant risk of contamination. Each apparatus should be monitored carefully for residual radioactivity before removal from the decontamination area and before use with other persons.

There is no reason to leave a wound open unless closure has been so long delayed that infection is likely. When drains are used, dressings should be monitored for radioactivity. Care should be used to collect any scab when it detaches from the wound. Often appreciable contamination can be found in it when it is analysed.

Since any débridement involves some risk of promoting the translocation and absorption of the contaminant, the surgical procedures for removal of long lived alpha emitters, such as plutonium or americium, should be performed while, or after, DTPA is given systemically. DTPA solution can also be used as an irrigation fluid.

In all cases of contaminated wounds, urine and faecal bioassay and/or in vivo body counting should be used to estimate the uptake of the contaminant into the body. All removed tissue, gauze sponges, and irrigation water should be retained for radiochemical analysis. Since it is seldom possible to remove all contamination during the initial decontamination effort, dressings should be monitored for contamination.

When an extremity is severely contaminated and adequate decontamination is not possible, the surgeon may be faced with the decision whether or not to amputate. Unless the extremity is so severely injured that functional recovery is unlikely, or unless the beta-gamma contamination is so severe that extensive and severe radiation induced necrosis can be expected, amputation is rarely indicated. Long lived alpha emitters such as plutonium may represent a potential long range threat, but conservative treatment should be tried first and the decision to amputate postponed until the potential long term risks are clearly defined and understood.

4.3.5. The fate of and tissue reaction to unremoved contaminants

As mentioned above, soluble compounds are often rapidly taken up into extracellular fluids and enter the metabolic pathways of the particular element and its compound. Insoluble compounds, especially if they are fine particulates, migrate along lymph channels to the regional lymph nodes. The regional node often is only a partial and temporary block in which case the contaminant moves on to the thoracic duct and enters the general circulation. For this reason, excision of a regional lymph node is seldom a definitive step in decontamination. It could be done, however, when monitoring reveals a large concentration in the regional node.

Depending on the type and amount of radiation emitted by the contaminating material, tissue reaction to intense local irradiation may occur and may influence migration of the contaminant.

It is extremely important to check all wounds for contamination when skin contamination has been detected. When mixtures of alpha, beta, and gamma emitting contaminants are possible, the alpha emitting contamination may be overlooked rather easily because of the ease with which the beta-gamma emitting contamination is detected. It is also necessary to consider the possible chemical hazards of the compounds along with the radioactive contaminants, as well as bacterial or other microbiological contamination.

When small amounts of contamination remain in the wound, the site should be monitored periodically in order to detect translocation. An inspection of the wound every year is advisable to identify nodule formation. If found, excision, followed by radiochemical and pathology studies of the nodule, is recommended.

4.4. INTERNAL CONTAMINATION

4.4.1. Principles of treatment

The procedures recommended for the treatment of persons with acute internally deposited radionuclides are intended to reduce the absorbed radiation dose and hence the risk of possible future biological effects. These aims can be accomplished by the use of two general processes: (i) reduction of absorption and internal deposition and (ii) enhanced elimination or excretion of absorbed nuclides. Both are more effective when begun at the earliest time after exposure.

Treatment is most effective if the uptake of contaminants into the systemic circulation is prevented. Administration of diluting and blocking agents is effective in some instances because it may also enhance the elimination rates of the radionuclide or reduce the quantity of radionuclide deposited in tissue. Therapeutic measures that use mobilizing agents or chelating drugs are less effective when the radionuclide has already moved into the tissue cells.

The most important considerations in treatment are: (i) selection of the proper drug for the particular radionuclide; (ii) timely administration after exposure.

In situations of substantial potential risk, resulting from internal contamination in which assessment of its level cannot be made readily, prompt institution of an effective therapy is clearly indicated. In other cases, for which assessment of the activity taken up and of resulting doses can be attempted on the basis of available data and/or relatively easy measurements, the intensity and methods of treatment should match the degree of expected risk. The treatment, involving sometimes procedures with inherent non-negligible risk, would perhaps be mandatory at expected organ doses leading to non-stochastic injury of clinical significance. If only stochastic consequences are expected, only simple, non-invasive procedures should be considered, if deemed necessary.

4.4.2. Reduction of gastrointestinal absorption

Gastrointestinal absorption can be reduced either by washing out or by the use of medications selected for specific radionuclides. These medications combine with the radionuclides so that absorption from the GI tract is reduced and the radionuclides are then eliminated in the stool.

Use of a nasogastric or gastric tube to wash out the stomach (*stomach lavage*) would generally be used in the highly unusual circumstance where the known oral intake of a high activity of radionuclides will pose a significant threat to the present or future health and where intake has occurred recently so that the material is still in the stomach. In such cases, stomach lavage would be the procedure of choice but may not always be successful. *Emetics* act by stimulating the gastric mucosa, by stimulating the vomiting centre in the brain, or by a combination of the two. Their use is contraindicated if the state of consciousness is impaired or after the ingestion of corrosive agents. Apomorphine hydrochloride and ipecac are the drugs to be considered for this use.

Of particular concern in the selection of a *purgative* is the residence time of relatively insoluble radionuclides in the colon and rectum. These portions of the gastrointestinal tract will receive the largest radiation doses but the damage can be reduced by prompt removal of the radionuclides from the intestinal tract. Some purgatives may have special advantages because they produce a less soluble compound of the radionuclide. Magnesium sulphate, for example, is a saline cathartic that can produce relatively insoluble sulphates with some radionuclides, for example, radium, and thus reduce absorption. Purgative drugs taken orally, such as bisacodyl, castor oil, or phenolphthalein, require several hours before taking effect. Use of *enemas* will empty the colon in a few minutes and may also be a consideration in some cases.

Activated charcoal would seem to be a potentially useful substitute but has yet to be tested extensively in this connection.

Ferric ferrocyanide (Prussian blue) has been found effective in accelerating the removal of caesium, thallium and rubidium by the faecal route. Prussian blue is essentially non-absorbed from the gastrointestinal tract and has low toxicity. It reduces the biological half-time of ^{137}Cs to a third of the usual value. (Radiogardase caesium is the commercial name for Prussian blue.)

Aluminium containing antacids are effective in reducing intestinal absorption of radioactive strontium.

Barium sulphate. The principal indication for barium sulphate in this application is as an immediate antidote for ingested strontium and radium. Formation of the insoluble sulphates of these elements will markedly decrease their intestinal absorption.

4.4.3. Blocking, diluting and displacement agents

A blocking agent saturates the metabolic process in a specific tissue with the stable element, thereby reducing the uptake of the radionuclide. Isotopic dilution is achieved by the administration of large quantities of the stable element or compound so that, on a statistical basis alone, the opportunity for incorporation and exposure of the radionuclide is lessened. Displacement therapy is a special form of dilution therapy in which a non-radioactive element of a different atomic number successfully competes with the radionuclide for uptake sites.

Protection of the thyroid from the effects of radioiodine is of concern not only for one or a few accidentally exposed individuals, but can even affect large numbers of the public.

To be most effective the blocking dose of stable iodide should be given as soon as possible. Recommended doses range from 70 to 300 mg iodine (100–390 mg of KI) (once per day), if it is decided that blocking is needed. (The biological half-time of stable iodine is 120 days, the effective half-life of ^{131}I (eight days physical half-life) is 7.5 days.) A comparable dose of potassium iodate can be used.

Concern has been expressed about the administration of stable iodide as having either damaging effects on the thyroid gland or systemic effects. For these reasons the risk–benefit relationship of administration of stable iodide should be weighed carefully. Intervention levels from 0.1 to 1 Gy to the thyroid have been proposed before stable KI administration is recommended [12].

The possibility of side reactions, although very rare, to milligram amounts of stable iodide, should be considered in the assessment of the benefit and risks of this type of treatment.

Phosphate may be useful as a *diluting agent* in case of medical misadministration of ^{32}P. Oral phosphates can be given in inorganic (sodium or potassium phosphate) and organic forms (sodium glycerophosphate). Vomiting, diarrhoea, or both, may occur from phosphate administration in doses exceeding 2 g per day.

In cases of very high exposure to tritium, a high level of fluid uptake by mouth will increase tritium excretion. Fluid forcing should be continued for at least 1 week or until further reduction in dose is limited. The half-time of tritium in the body can be reduced from the normal ten to twelve days to five days or less by forcing at least three to four litres of fluid per day. The radiation dose may thus be reduced by a factor of two or more by careful management.

As an example of a *displacement agent* given orally or intravenously is the administration of calcium to increase the urinary excretion of radioactive strontium and calcium in man.

4.4.4. Mobilizing agents

Mobilizing agents are compounds that increase a natural turnover process, thereby enhancing elimination of radionuclides from body tissues. These agents are more effective if they are given soon after exposure, but some still produce an effect if given within about two weeks.

Chelating agents may be considered to be a special class of mobilizing agents and are discussed in Section 4.4.5.

When radioactive iodine has already been deposited in the thyroid, treatment with stable iodine is not effective. Antithyroid drugs such as propylthiouracil and methimazole might be considered in these cases if the radioactive dose is high enough to justify the use of these drugs with their potentially dangerous side effects.

Ammonium chloride, given orally, is effective in mobilizing radiostrontium deposited in the body. Its effectiveness can be enhanced by simultaneous use of intravenous calcium gluconate.

Diuretics are untested for the treatment of internal radionuclide deposition with the exception of tritium. Enhanced excretion of sodium chloride, potassium bicarbonate, magnesium, and water in the urine occurs with induced diuresis. Some corresponding radionuclides that could be associated with radiation accidents are ^{22}Na, ^{24}Na, ^{38}Cl and ^{42}K.

Apart from DTPA, studies on the effect of expectorants and inhalants on inhaled radioactive particles have been disappointing. None provides effective action that would be dependable or particularly useful in treating persons after the inhalation of radioactive particles. There is need for further scientific study to confirm the initial studies with animals.

4.4.5. Chelating agents

A number of chemical compounds enhance the elimination of metals from the body by chelation, a process by which organic compounds (ligands) exchange less firmly bonded ions for other inorganic ions to form a relatively stable non-ionized ring complex. This soluble complex can be excreted readily by the kidney. A properly selected and administered chelating drug will enhance the excretion of some radionuclides and thus reduce their residence times in the body. Therapy with a chelating agent is most effective when it is begun immediately after exposure while the metallic ions are still in circulation and before their incorporation within cells or deposition in bone.

The calcium salt of ethylenediaminetetraacetic acid (*calcium edetate*, CaNa EDTA or CaEDTA) is the most common form of chelator used in man, primarily to treat lead poisoning. It can also be used to chelate zinc, copper, cadmium, chromium, manganese and nickel. It has some effectiveness for the transuranium metals, such as plutonium and americium, but $CaNa_3DTPA$ has been found to be more effective by an order of magnitude for those radionuclides.

The edetates are nephrotoxic and must be used with extreme caution in patients with preexisting renal disease. Transient bone marrow depression, mucocutaneous lesions, chills, fever, muscle cramps, and histamine-like reactions (sneezing, nasal congestion, and lacrimation) have also been described.

The powerful chelating agent diethylenetriaminepentaacetic acid (pentathamil, DTPA) is generally more effective in removing many of the heavy metal, multivalent radionuclides than CaEDTA. It is effective for the transuranium metals (plutonium, americium, curium, californium, and neptunium), the rare earths (cerium, yttrium, lanthanum, promethium and scandium), and some transition metals (zirconium and niobium). The clinical use of Ca- and ZnDTPA has been primarily for treatment of plutonium and americium exposures.

CaDTPA also binds trace metals present in the body, such as zinc and manganese. It is the reduction in these two trace metals that probably accounts for the toxicity to high doses in animal experiments.

The zinc salt of DTPA is less toxic than CaDTPA and therefore is advantageous to use for longer term treatments and especially for fractionated treatments.

CaDTPA is more effective than ZnDTPA in rats when given promptly after exposure to ^{239}Pu, ^{252}Cf, or ^{241}Am. This finding led to the general recommendation that CaDTPA be used during the first 24 to 48 hours after exposure and then that ZnDTPA be used for continuing treatments.

The effectiveness of DTPA in enhancing the excretion of plutonium is markedly affected by the chemical form of the plutonium. For both wounds and inhaled particles, the uptake of relatively insoluble plutonium compounds, such as plutonium oxide, into the circulation occurs over many days and weeks. DTPA is not effective in these cases because of the small amount of plutonium present soon after exposure in the blood or intracellular fluids. Soluble compounds, such as plutonium nitrate, have relatively rapid uptake and translocation, and so the plutonium is more available early after exposure for chelation. Data from persons treated with CaDTPA soon after exposure (on the first day) indicate that about 60 to 70% of the soluble forms of plutonium are removed.

Both CaDTPA and ZnDTPA are available as an investigational new drug in the United States and can be obtained through the Radiation Emergency Assistance Center and Training Site, Oak Ridge, Tennessee. They may also be readily available from pharmacies, as is CaDTPA in the Federal Republic of Germany. These drugs have been administered by both intravenous injection and aerosol inhalation.

In cases of uranium incorporation, chelating agents should not be given since the kidney may then be subjected to uranium overload. Treatment to remove uranium intakes is not particularly successful, but sodium bicarbonate in saline may be given by slow intravenous infusion.

There is evidence that the water soluble derivative DMPS (*dimercaptopropansulphonate*) available as *Dimaval* in many countries[2] is more effective and less toxic than BAL (British Anti-Lewisite, i.e. dimercaprol). Its use has been recommended for treatment of mercury, lead or polonium incorporation.

Penicillamine, an amino acid derived from the degradation of penicillin, chelates with copper, iron, mercury, lead, gold, and possibly other heavy metals. It is superior to dimercaprol and CaEDTA for the removal of copper.

Deferoxamine (DFOA) has been used effectively in the treatment of iron storage diseases and acute iron poisoning. If given promptly, DFOA surpasses CaDTPA in the enhancement of excretion of plutonium (IV) compounds. Its effectiveness declines rapidly, which makes its clinical use for this purpose questionable. The combination of DFOA and CaDTPA yields better results than either drug separately.

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[2] Available, for example, in the People's Republic of China, the Federal Republic of Germany, Poland and the USSR.

4.4.6. Contamination of respiratory tract and lung lavage

Deposition of radioactive particles in the lung is one of the more common types of accidental exposure of humans to radionuclides. Insoluble particles, once inhaled into the lung, may be mobilized and translocated to other organs at a low rate over many months or years.

The lung normally clears inhaled products to varying degrees. One of the pathways for clearance is the digestive tract, following the return of inhaled material through the bronchial tree, secondary deglutition and passage into the oesophagus. Clearance via the digestive tract may in fact involve a large fraction of the total contamination — for example in the case of particles of relatively large diameter. As a result, there may be secondary digestive absorption, depending on the nature of the inhaled radionuclide. It is thus essential, after an accident involving serious contamination of the lungs by a radionuclide which is also absorbable in the digestive system, to try to make it insoluble in the gastrointestinal tract. In the case of beta emitters, exposure of the mucosa can be reduced by using laxatives. When gaseous contaminants are involved, no therapy is available except in the case of gaseous radioiodine, when immediate blocking of the thyroid with stable iodine may be indicated.

After the inhalation of relatively soluble radionuclides, lavage of the tracheobronchial tree has proved effective as a treatment technique in a very limited number of cases.

Possible use of this experimental technique in man requires a careful risk-benefit assessment. The risk lies primarily in the administration of a general anaesthetic. Thus, this procedure should be considered only in high exposures in which a reduction of the dose could be expected to prevent acute or subacute effects, such as radiation pneumonitis or fibrosis.

4.4.7. Enhancement of elimination by extracorporal treatment techniques

This method used in medical toxicology can be extended for the removal of radionuclides or labelled compounds while they are circulating in the blood stream. Depending on the chemical properties and metabolism of the radionuclide compound either haemodialysis or haemoperfusion is used. In haemoperfusion the blood is led directly over a column of activated charcoal or resin. In haemodialysis the blood is filtered through a dialysing membrane.

CHAPTER 5

ORGANIZATION, PLANNING AND TRAINING

5.1. INTRODUCTION

The efficient management of radiation accidents depends on the *planning* and outfitting of the facilities, and the *training* and organization of their personnel. This is particularly true for first aid establishments with responsibilities for caring for injured individuals. National authorities should set up a system for handling radiation accident situations geared to the real need of the country. Preplanning by public health authorities to provide an adequate medical radiation emergency response should consider the use of existing facilities in specialized hospitals to be adapted in case of need to deal with radiation emergencies. Hospitals equipped with burn centres, haematological departments and individual care units can easily be adapted to cover external irradiation emergencies. It is, however, quite difficult to maintain high levels of readiness and efficiency because dedicated radiological safety practices successfully reduce the incidence of radiation accidents by many orders of magnitude below that of heavy industry at large. Organization, planning and training must therefore be tested by well planned drills held at unpredictable times which address the potential problem of a specific site of some specific plant.

In this chapter suggestions are made concerning organization, outfitting, planning, training and drilling.

5.2. THE PROVISION OF DIFFERENT LEVELS OF ASSISTANCE

Radiation accidents may involve the irradiation and/or the contamination of individuals. The medical assistance to be provided may be considered at several different levels depending on the seriousness of the accident. These levels would extend from a local on-site emergency station to a large central hospital with specialized facilities.

This section offers guidance on the measures that should be provided for accidentally irradiated individuals or contaminated persons. It describes four increasingly sophisticated levels of assistance. The following section (5.3) provides additional information on the special requirements for assistance in accidents involving radionuclide contamination.

5.2.1. On-site emergency assistance

On-site emergency assistance relates to cases of *accidental* radionuclide contamination, external and internal, since an *irradiation* casualty does not require any emergency treatment. If emergency treatment is considered necessary when there is a suspicion of internal contamination, such as cases involving the transuranic elements, this calls for implementation of treatment even before the physician arrives, in order to reduce to the minimum the time between contamination and treatment. It therefore may be desirable that anybody exposed to any risk of *contamination* with such radionuclides should have a first aid kit of compact size and containing a small selection of simple, non-toxic medicaments put up in individual dose units and intended to be taken by the contaminated patient himself at the site of the accident. This first aid kit can be used only if the staff is properly informed and trained. Therefore education and instruction must be one of the main tasks of the medical service.

The safety regulations for the workplace should cover the use of the kit on the merest suspicion of internal contamination by highly radiotoxic nuclides. The harmlessness of the treatments prescribed, in view of the quantities of medicaments available in the kit, should make possible a regulation of this kind without contravening the law. Naturally, the use of the personal kit does not under any circumstances excuse the contaminated person from reporting to a physician whose job it is to confirm the contamination or not, to follow up the treatment as necessary, and to make up a fresh kit.

5.2.2. Facility medical service

This service constitutes the second stage for the casualty. Whereas first aid as described above may not be appropriate, nevertheless the medical service responsible must be involved. The means at its disposal will depend on the size of the facility and the hazards incurred there. The ideal is a medical service on the site of the facility itself, having the resources necessary for isolation, decontamination and rapid measurement of external and internal contamination of casualties, in addition to premises suitable for these purposes. The service must possess equipment and laboratories enabling it to handle these casualties in an efficient manner, i.e.:

(a) To perform rapidly the necessary sampling and biological examinations;
(b) To treat several contaminated persons at the same time without spreading the contamination (i.e. availability of rooms separated by airlocks for undressing and others for showers or baths, and of a one-way system ensuring that a decontaminated individual does not go back into a suspect area);
(c) To provide activity measuring equipment adapted to the hazards involved, ranging from a portable detector to a whole body counter;
(d) To ensure that the facility has a list with names, phone numbers and addresses of specialists who may be called for assistance;
(e) To provide — and this is most important — the medicaments needed for treating these particular casualties. In small or medium size medical services

it is desirable to have first aid kits similar to those for individual use on the site of the accident, but intended for the physician himself. Such *group medical kits* may contain, for example, seven daily doses for treating one person for seven days or seven contaminated persons for one day. The purpose of this kit is to assemble in compact form all the non-current medication that should be available when a contaminated person is received in a medical unit. The aim is to simplify the action that needs to be taken, so that the medical requirements can be fulfilled promptly.

(f) In order to protect medical and allied personnel who are responsible for decontamination and medical care of a highly radioactive patient, it is necessary to perform continual monitoring with detectors of appropriate sensitivity in order to limit the contamination of the personnel.

The small size of many nuclear facilities does not justify a medical service with such resources and a large staff. It is here that an accident could present most problems. In large facilities the first two stages in treatment — emergency treatment and treatment at the facility medical service — can be combined in time and space, normally without any undesirable effects for the casualty, but in a small facility with only a small medical service this is more difficult, especially if the medical service is situated away from the site. In this case the physician in charge, who would not have at his disposal large scale resources for the purpose of making a preliminary assessment, can only refer the casualty to the nearest or best equipped hospital. In every case transport must be adapted to the person's condition. Whereas an irradiated individual can be evacuated without particular precautions, in the case of a contaminated patient it is necessary to take steps to avoid spreading the contamination, especially if in the facility no external decontamination has been performed. Certain items of special equipment, such as stretchers with isolation sheets that are airtight and adaptable to various uses, ambulances lined inside with disposable plastic, etc., can for the most part solve this problem and, moreover, make it possible to transport the contaminated individual even when he is also injured.

5.2.3. Regional hospital

The essential role of the regional hospital is to make up for the inadequacies of the local medical service. The regional hospital must also be provided with a list of experts in a variety of disciplines (surgery, haematology, internal dosimetry, etc.) who can be called in for consultation or assistance. The names, addresses, phone numbers, etc., have to be incorporated in the overall emergency scheme. It may be essential, on surgical grounds for example, to hospitalize the patient urgently, and in this case only a regional hospital can cope with the situation. It may also happen that it is preferable for social, economic and/or psychological reasons not to move the patient. This is the case where, for example, a contaminated wound requires only a simple surgical intervention; in this case the operation can simply be done with the attendance of a specialist in radioprotection. It is this specialist who comes to the patient with his measuring equipment.

Thus, although it is desirable to reduce the involvement of the regional hospital, the latter should be in a position to deal with a certain number of situations. Although vast resources will not usually be available to provide equipment at this level, e.g. isolation chambers and operating theatres designed from a nuclear viewpoint, and built and maintained against the eventuality of an accident, it is none the less necessary to make a survey of existing resources and to adapt them to specific needs. Thus it is, for example, preferable to have a special entrance for contaminated patients, with direct access to a suite of rooms that can be adapted for purposes of decontamination and are immediately available. Standard operating theatres equipped for aseptic conditions are suitable for typical operations, and effective external decontamination of the casualty makes it possible to avoid spreading radioactive contamination to any extent.

5.2.4. Central hospital

It is at the level of the central hospital that the large scale medical treatment resources are to be concentrated. One central hospital is sufficient for a country of medium size and might cope with accidents occurring in several countries. There are two conditions governing the design of large scale resources to be used in cases of irradiation or contamination: (a) these resources should be used throughout the year for other, routine purposes, on grounds both of economic viability and of reliability; (b) they should be immediately available in the event of an accident.

A central hospital may be capable of dealing with the consequences of both irradiation and of contamination accidents, although the resources used in each case are very different. The two types of case can be handled efficiently only if there is close co-operation with specialized laboratories, some of which may not have the usual relationship with the central hospital, i.e. it is important at this level that the various measures to be taken should be effectively co-ordinated, final decisions naturally remaining the responsibility of the chief physician of the central hospital.

Unfortunately, in many countries the existing hospital infrastructure does not provide for such a service, and the procedures adopted must be based on what is available. In the case of an irradiation accident, the choice of the appropriate procedure, and hence of the medical service involved, is relatively limited. Resuscitation units and units for severely burned casualties are among those which provide the optimum conditions for the patient; the staff of such units is familiar with all techniques for isolation and asepsis. Confinement in a sterile atmosphere, if it is decided that this is necessary, can be achieved by setting up a light plastic isolation tent, which has the advantage of being immediately disposable and portable and making it unnecessary to move an ordinary patient in order to permit treatment of the irradiated individual. The specialists involved are numerous and include members of the dosimetry team. Although in most cases the latter have no formal link with the hospital, they must carry out their functions as soon as possible. Apart from the preliminary results, which indicate an order of magnitude of the irradiation, reconstitution of the accident provides information on the spatial dose distribution, which is a vital element in prognosis.

A contaminated patient presents particular problems. A nuclear medicine or radiopathology service can handle such a situation without undue adjustments having to be made, as it will possess measuring equipment and facilities to store and dispose of radioactive effluents; it will have a staff that is used to the problems presented by the handling of radionuclides. However, the special conditions attending surgical intervention on a contaminated individual must be met by co-operation between the surgeon and the specialist in radioprotection.

In general, the central hospital, however sophisticated it may be, does not have on its staff all the specialists needed for a radiation accident — whether involving internal contamination or external irradiation - since the disciplines called for are numerous and sometimes go beyond the medical sphere. Co-ordination of these numerous specialists, each of whom has different responsibilities, is essential, even if only to draw up an order of priority for the measures to be taken. It is with this in mind that relations between the medical service of the facility and the central hospital must be maintained, and the latter must be regularly kept abreast of the particular problems posed by casualties involved in radiation accidents.

The number and type of specialists involved and the resources employed must be adapted to each individual case. Thus, there is no strict rule governing evacuation of the casualty; it is clearly preferable to evacuate the person to the specialized unit in which he will find the treatment best adapted to his condition. However, it may be that for medical, logistic or psychological reasons the best solution is to minimize the transport of the casualty and to keep him in the local hospital. In this case it is the specialist team, accompanied by the necessary equipment, that comes to the bedside of the patient.

5.3. ADDITIONAL CONSIDERATIONS FOR ACCIDENTS INVOLVING RADIONUCLIDES

5.3.1. Organization and planning of a first aid station

The complexity of the organization of first aid stations depends on the number of people employed, the particular hazards of the workplace and the potential seriousness of the radiation accidents. The first step therefore in the organization of a specific first aid programme depends upon an intimate knowledge of the operations of the plant and the radionuclides involved. Such an analysis must also include the identification of geographical and transportation problems in relation to the size of work forces at satellite sites. From this the number and location of first aid stations and their staff and equipment needs are determined. The station and staff then are integrated into the medical management of the central plant dispensary. The complexity of the communication network is defined by the need for backup support between individual first aid stations as well as by the dependence of their staff upon central dispensary direction. A redundant communication system in an accident response plan, where the central and outlying sites are integrated, makes it unnecessary for all sites to have the same capabilities for responding to the most serious accident that can be foreseen at the plant or to be equally equipped. The response plan to be developed starts therefore with an analysis of the hazards in the immediate territory of a particular first aid station and plans are tailored individually to meet them. The plans should then provide for increasing levels of inter-site support and all-facility response as the seriousness of an accident and the number of people involved increase beyond the capabilities of one or several first aid stations.

Even though a small plant or laboratory may not require a full time physician or ancillary medical staff, it will need a consultant physician, a well trained first aid team and an experienced radiological safety officer. In the event of an accident, the first aid team could handle immediate first aid while contacting the consulting physician who would direct the next actions. Based on a previously established plan, the patient could be taken to an area hospital which has a decontamination facility and a trained medical staff, or he could first be transported to another, larger and better equipped in-plant decontamination facility or occupational medical department and later to the hospital if necessary.

Preplanning is obviously of paramount importance. Knowing the potential radiation hazards in a particular plant or laboratory and having a careful action plan designed to handle all credible accidents is far more important than elaborate and expensive facilities.

The basic first aid team member is a physician trained in the management of radiation accidents. The other first aid team members are radiation workers who have received special first aid training, and/or trained nurses and/or technicians. One or several health physicists familiar with radiation detection instrumentation and the special radiation protection problems associated with accidents are also essential. A designated central manager, not actually involved in administering first aid, must be available to perform the valuable service of co-ordinating communication, technical support, transportation and management of personnel in the accident area who have not been obviously contaminated or exposed, but who require careful monitoring for radioactivity.

5.3.2. Facilities and equipment

A minimum first aid facility should have an area where both decontamination and first aid can be performed. It may consist only of a room painted with strippable paint, equipped with an autopsy table with a central drain, and hot and cold water which can be mixed and delivered in a flexible plastic or rubber tube with a shower head. Alpha emitting radionuclide contamination requires no shielding, but inhalation and skin contamination of team members must be avoided. Elaborate decontamination facilities have been designed and contructed for large plants, but some are probably overdesigned and unnecessarily expensive.

The waste water from the decontamination table should be collected in a holding tank for later monitoring and special disposal as required. Special ventilation and air conditioning may be necessary to prevent air contaminated with alpha emitting radionuclides from flowing to other rooms. The members of the first aid team must have effective respiratory protective devices that have been prefitted so that these personnel are protected from inhaling radionuclide contamination which may become airborne during the decontamination effort. Full protective clothing, including coveralls, shoe covers, caps and plastic gloves, should be put on by the members of the first aid team before they arrive on the site of an accident; it may have to be changed if it becomes excessively contaminated during the initial decontamination effort.

If possible, the patient should have all contaminated clothing removed at or close to the site of the accident. If he is ambulatory, he can shower and wash himself under monitoring supervision. The patient is covered with sheets or blankets if it is necessary to keep him warm. If the initial decontamination cannot be completed, a plastic bag or sheet is used to prevent spread of contamination to the transporting vehicle, e.g. the ambulance. If the weather is warm, it should be remembered that a patient placed in a plastic bag can be subjected to considerable heat stress when transported 30–70 km in an ambulance.

A well equipped first aid kit with splints, bandages, tourniquet, life support medications, airway, basic emergency surgical instruments, intravenous fluids, etc. should be provided at all designated sites and periodically checked to ensure that all supplies are in good functioning order and sterile if necessary.

Each first aid facility should have plastic bags for collecting and labelling clothing, jewellery, etc., containers for collecting urine and faeces, and specimen bottles with formalin if freezing facilities are not available nearby. A portable tape recorder, enclosed in a plastic bag, with the microphone suspended near the decontamination table, can be helpful to record details of the history of the accident, physical findings and details of the decontamination. Several extra pairs of large bandage scissors should be included in any first aid kit, since clothing may need to be cut off the patient.

5.4. HOSPITAL PREPAREDNESS

Preplanning between the medical service and the hospital unit is necessary to assure that the latter is prepared for the special problems posed by a contaminated or irradiated patient. The extreme case, which should be avoided by such planning, is for a hospital to refuse to admit a person on the grounds of wishing to avoid contamination of its premises and staff.

As well as assuring that the hospital is provided with the necessary scientific and technical information, it is important that the particular problems associated with the nuclear nature of an accident are put in their proper perspective with respect to medical and/or surgical problems; moreover, one should not forget the psychological aspect often attached to anything involving the 'radiation mystique'. These secondary but important aspects are reduced to a minimum if the hospital service is trained in the handling of radioactive sources as is the case, for example, with a nuclear medicine service.

5.4.1. External irradiation accidents

In the case of a person exposed accidentally to radiation no measures are necessary to protect the premises or personnel. Since cases of whole body exposure are rare and there is no urgency in the medical sense of the term, there is no point for all hospitals in permanently setting aside a room for the admission of possible victims of radiation accidents. However, in view of the rarity of such cases and the possibility of their being serious it is essential that a plan be drawn up and adhered to. In particular, it is necessary to co-ordinate fairly frequently the actions of the various specialist services involved. These specialist services are numerous and often far removed from each other, both as regards the nature of their normal activities and their geographical location, and it is essential that continuous liaison should exist between them. Thus, to name only the principal ones, it will be necessary — on the clinical side — to be able to call upon specialists in haematology, toxicology, radiology, nuclear medicine, dermatology, neurology, gastroenterology, cardiology, dentistry and otolaryngology, as well as a reliable resuscitation unit and a wide range of biologists from fields such as pathology, with its highly specialized forms as, for example, cytogenetics, bacteriology, immunology, cellular kinetics and histopathology.

The best hospital service for the care of a whole body exposure victim whose prognosis is uncertain is one where the departmental staff is familiar with all the methods of isolation and asepsis. Such departments should be identified and used for preplanning and training. A major problem is whether or not to confine the patient in a sterile room. As it is impractical that a room should be permanently reserved as a sterile room and as, moreover, it might not always be possible to make such a room available for an exposure case by displacing another patient, it would seem preferable to use a light plastic isolation tent, kept folded away and regularly inspected, with its own independent filtration system and preferably maintained at overpressure while in use. Such a tent can be erected quickly inside any ordinary hospital room. The choice between sterile isolation and simple isolation against exogenous infections (infections of dental origin should not be overlooked here) is left to the specialist or team of specialists taking charge of the exposed patient. In any case, apart from specialized personnel, the hospital should have some special facilities available, including at the very least a means of isolating the patient.

Although an isolated patient should not as a general rule leave his place of isolation during his illness and it is rather the various specialists who should come and visit him, it is desirable to release the patient to his family at home until changes in his blood values suggest that further hospital care will be needed (about the 21st day). As it is clearly not desirable to expose a patient with agranulocytosis to the attacks of microorganisms which he is bound to encounter in moving about, it is a good plan to have individual sealed transport systems such as stretchers covered with a transparent plastic bubble. Provisions should be made for the isolated patient to have an audiovisual contact with family and friends, to prevent psychological pressures.

5.4.2. Contamination accidents

The contrast between a case of exposure and one of contamination can be readily understood when we consider the measures that have to be taken in a hospital. Whereas in the case of overexposure, efforts are directed towards protecting the victim, in the case of a contaminated patient additionally varying measures have to be considered for protecting the staff and the premises and sometimes the public as well. Moreover, an externally irradiated patient does not generally constitute a medical emergency; on the other hand, the prognosis of a contaminated patient depends most often on the speed with which treatment is commenced.

5.4.2.1. Measures for dealing with contamination

Admission to hospital does not generally pose any special problems, because in the majority of cases the external contamination of the victim has been treated and there is no substantial risk of its spreading. In exceptional cases the patient may have come directly from the place where he was contaminated (case of a person both injured and contaminated, for example); in this case it is desirable for the patient to be admitted by another route than that normally used by patients. Except in cases of medical emergency the patient should be taken to a shower room or to an autopsy table, i.e. a place where decontamination will be facilitated and spreading of the contamination within the hospital avoided.

The risk of staff being contaminated is very small in the majority of cases. Particular attention should be paid to residual contamination of the skin which can be spread during desquamation, or contamination of a wound which can be spread when the scab is shed, as well as to contamination which could be caused by excreta. The use of disposable gloves and disposable plastic containers reduces possible risks.

Special vigilance is called for in the case of an unconscious patient who is incontinent or vomits in the period following contamination of the GI tract, as well as in the handling of dressings placed on a contaminated wound. Not only should the usual precautions be observed in handling these dressings but arrangements should be made for keeping and labelling them for subsequent examination to establish the level of contamination.

Ideally, apart from the equipment usually available in a hospital, the following facilities should be available:

An isolated room that can be used for external decontamination of the patient, with a shower and an ocular douche;

Measuring equipment (for alpha, beta, gamma and X-radiation), capable of detecting low level contamination, that can be used for the skin and clothes, the ground and walls;

Containers for taking biological samples for radioanalysis;

A sufficient stock of sheets, towels, masks, gloves, general and surgical instruments, and plastic bags for storing contaminated solid waste.

Clearly, these facilities and equipment can be of help only if a minimum essential number of staff knows how to use them and has thus received appropriate instruction or been provided with all the necessary basic information.

To avoid loss of time, which could have serious consequences in the case of personal contamination, it is essential that informal or, better still, formal arrangements exist between the facility medical service and the hospital unit and that the persons working together know each other well.

5.4.2.2. *Measures to protect personnel against irradiation from a contaminated person*

So far, there has never been a case where the contaminated patient has been a radiation hazard to hospital personnel. If such a case should arise it would be necessary to cut down the time spent on both medical and general treatment of the patient and to check that the sum of the doses received during the time spent near the patient and/or from his excreta is less than the relevent dose limits. It is even more unlikely that recourse to the use of screens or markers (zone delineation) will be necessary. Isolation of the patient in a room with one bed may be advisable.

5.4.2.3. *Measures for dealing with waste*

Contaminated waste should be dealt with in accordance with the generally accepted regulations for handling radioactive waste.

As far as excreta are concerned, the problem is not one with which the hospital must concern itself directly since radiotoxicological tests are performed throughout the patient's stay in hospital and the laboratory concerned is used to dealing with the problem of waste. On the other hand, the hospital service may have to dispose of various liquid or solid wastes. Any suspect waste should be considered as a contaminant, as long as no proof to the contrary exists. Putrefying material such as faeces and urine should be stored in a freezer. When the radioactive contaminant has a short half-life, it is a good idea to keep it for the necessary period in a locked, ventilated room with a warning on the door. With certain highly contaminated materials it is often better to dispose of them as soon as authorized than to undertake a long, difficult, costly and incomplete decontamination process.

5.4.2.4. Measures to be taken in the event of a surgical procedure

The measures to be taken in the event of surgical procedures on contaminated wounds and pulmonary lavage will be dealt with in a future publication. Only the special precautions intended for protecting the hospital and its staff will be recalled briefly here:

Protect the operating theatre, covering the table, floor, etc., with disposable plastic sheets;

Provide detection instruments appropriate to the nature of the contaminant, some for monitoring the operative fields, surgical instruments and gloves, and others for monitoring the theatre and waste;

Monitor the respiratory or digestive probes in the case of pulmonary or gastro-intestinal contamination. Fit the mucous extractor with a catch bottle;

Use disposable materials as much as possible and consider as waste all contaminated clothing worn by the surgical team.

5.4.2.5. Measures to be taken in the event of death of a contaminated person

It is difficult to formulate general rules concerning the autopsy, burial or cremation of a body containing a certain amount of radioactive material, because the laws on these subjects vary from one country to another. Storing a body for several days at -20°C or -30°C helps to overcome the problem when short lived emitters are involved. It is in fact rare for the contamination level to be such as to pose problems for burial. On the other hand, it is necessary to consult the regulations if the body is to undergo special preparation, to be cremated, or to be embalmed. This practice should be avoided unless it simply involves injecting a fixing substance. Nevertheless, the persons performing the embalming should take all the usual precautions for handling radioactive contaminants.

The same precautions should be taken during the autopsy, which should be made as brief as possible if the contamination level is high. In the latter case the pathologist should be assisted by a radiation protection officer.

5.5. TRAINING AND DRILLING

Since radiation accidents are relatively rare, personnel who will have to handle the first aid care cannot expect to derive their skill from experience. They must receive careful training at formally held courses, by practice drills and by taking on responsibilities for inspection and review of their own facility and, if possible, other production or research facilities. The results of the drills should be recorded and kept, including records of first aid exercises.

Practice drills, if realistically staged with a carefully developed scenario, can identify deficiencies in the preparations for an accident. After a drill, a critique should always be held. Experienced observers can then advise all participants of their mistakes or lack of judgement and skill. Revisions in plans and upgrading of skills, which should occur after each exercise, help prepare for the real accident, if and when it occurs.

Inspections of production and research facilities, at least by senior members of the first aid team, enable the team members to challenge their supervisors to speculate on possible accidents. They learn what radiation devices, sources or radionuclides are being used and how an accident might occur. Armed with the knowledge of the possible accident, the preparation for handling it should be relatively easy. Members of a team can be required to develop written plans for accident responses which can then be reviewed for completeness and clarity.

CHAPTER 6

MEDICAL ADVICE FOLLOWING ACCIDENTAL OCCUPATIONAL EXPOSURE

Any accident involving internal contamination or external exposure requires certain decisions to be taken by the physician; as some of these decisions can have important consequences for the worker concerned, all the factors must be weighed very carefully. In all cases lessons should be drawn from the accident and steps be taken to improve plant operation and protection of workers. To this end, the investigation of the accident should:

(a) Examine the circumstances of the accident in consultation with the radiation protection staff and those responsible for the operation of the facility in question;
(b) Record all the results of measurements carried out, noting carefully the dose to organs or tissues, as well as decisions of a medical and/or administrative nature that have been taken as a result of dose limits having been exceeded.

In addition, the physician should conduct follow-up examinations on such persons to determine whether or not the accident has physical or biological sequelae. Any change of a medical nature occurring some time after the accident should be recorded.

The physician should ensure that the worker concerned is fully informed of the situation. In particular he should inform him of all the risks to which he is subject as a result of his contamination or exposure, this being especially important when the worker is returning to his former job. Thus, in the case of a job involving the risk of further contamination or exposure the physician should explain clearly in terms comprehensible to the worker the pathological effects which could result in the long term. Talking to the worker in this way is also likely to make him take extra care so as not to allow the accident to happen again.

The administrative decisions which the physician has to take are very often complicated. Following a radiation accident and discussion with the physician, in the course of which the significance of his exposure has been explained, the worker should in general be encouraged to resume his occupation. However, attention should be paid as to whether he was personally partly to blame for the accident. This could then be a decisive argument for moving him to another job.

In the majority of cases, management in consultation with the physician and the radiological health and safety officer must take into account non-objective criteria such as socioeconomic status, age, sex and job expertise to arrive at a decision on the patient's job fitness. The problem is not just to evaluate the probability of sequelae but to evaluate the possible risks in the far distant future. The management team must base its decision on several considerations: knowledge of the risk with all the uncertainties which this involves, knowledge of the problems of reclassification for the person concerned, and knowledge of the psychological problems which a decision of unfitness for work could result in for the worker, his colleagues and his family. However, the physician himself should decide on the frequency and nature of the medical examinations the patient should undergo, taking into account the nature and the gravity of the accident.

Particular problems arise when the embryo or foetus has been irradiated beyond a small number of millisieverts. In this case specialist advice should be sought.

Appendix 1

INTERNATIONALLY AVAILABLE RADIATION ASSISTANCE PROGRAMMES

Since 1959 the IAEA has had an action plan to arrange for assistance after an accident involving radioactive materials. There are three main aspects to the current plan:

(a) Member States are encouraged to determine in advance what outside assistance would be needed in the event of an accident and then to enter into a multilateral assistance agreement with neighbouring states. In 1963, the "Nordic Agreement" was signed by representatives from Denmark, Finland, Norway, Sweden and the IAEA.

(b) Through training programmes Member States have been encouraged to develop their own capabilities to handle emergencies. Such programmes have been conducted in Manila, Tehran and Buenos Aires.

(c) The IAEA is prepared to arrange for specialized assistance after an accident, e.g. for medical and radiological support. Starting in 1963, the IAEA has issued periodically an internal report on Mutual Emergency Assistance for Radiation Accidents which identifies what assistance can be made available at the request of another state. WHO, FAO and ILO have participated in later revisions of this publication; the most recent was issued in 1971 [13].

The IAEA maintains the capability to have a senior technical person available through a duty officer roster in the event of a telex or telephone request for assistance. The programme also includes the capability to send a small group of observers or consultants to the site. The facilities of the Agency's laboratory for radiochemical analysis of environmental samples and for bioassay and whole body counting are also available.

In April 1977 the IAEA entered into an agreement with the United Nations Disaster Relief Organization (UNDRO) by which technical support will be provided in the event that disaster relief involves radiological aspects. In addition, the World Health Organization has recognized the Hospital of the Institut Curie in Paris, the Oak Ridge REAC/TS facilities and the Leningrad radiobiology centre as 'collaborating centres' for the purpose of giving assistance in special cases of accidentally exposed individuals.

Appendix 2

CENTRES FOR TREATMENT OF ACCIDENTALLY EXPOSED INDIVIDUALS AND FOR TRAINING

1. REAC/TS: ITS ROLE AS A SPECIALITY REFERRAL CENTRE AND TRAINING SITE

In 1976 the US Department of Energy established REAC/TS, the Radiation Emergency Assistance Center and Training Site, as a part of the Medical Division of Oak Ridge Associated Universities (ORAU). In the 40 months of operation since then, REAC/TS and its programmes for radiation accident response, radiation accident management training courses, accident registries, cytogenetic dosimetry, and diethylenetriaminepentaacetic acid (DTPA) distribution and therapy has established itself as an internationally recognized centre for all phases of radiation accident management. In this appendix the current role which REAC/TS plays in emergency medical response to radiation accidents, as well as its training activities conducted within the facility are described.

The speciality referral centre concept

Modern, state of the art emergency medical response sometimes requires personnel and services unavailable at many community hospitals and major medical centres. Such services are well defined, and need highly specialized equipment and a staff trained in all aspects of emergency or intensive care service. This specialization has led to the establishment of referral centres which provide these unique services. The speciality referral centre concept is illustrated in Fig. 6. In the majority of situations, a speciality referral centre is physically attached to a local hospital, thereby facilitating easy transfer of patients. REAC/TS, located within the Oak Ridge Hospital of the Methodist Church in Oak Ridge, Tennessee, is such a speciality referral centre in the United States for the treatment of radiation accident victims. It is only one of many such centres in the continental United States. The Department of Energy maintains a number of such installations including the Hanford Environmental Health Foundation, Richland, Washington, and the Savannah River Plant at Aiken, South Carolina, and medical facilities at the Brookhaven National Laboratory, Upton, Long Island, New York, and at the Los Alamos National Laboratory in New Mexico. Other radiation medical centres are found in the major hospitals of the Department of Defense and in the Clinical Center of the National Institutes of Health in Bethesda, Maryland. Many university medical centres have expertise in the management of radiation accidents, such as the University of Pittsburgh Medical Center, the University of Cincinnati Medical

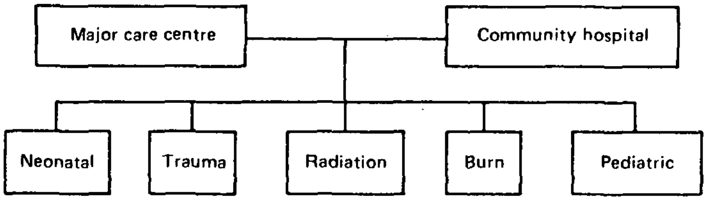

FIG. 6. Block diagram illustrating the speciality referral centre concept and the interrelationship between these centres and conventional hospitals.

TABLE VI. SUMMARY OF ASSISTANCE AGREEMENTS ESTABLISHING REAC/TS AS A MEDICAL BACKUP FACILITY FOR RADIATION ACCIDENT MANAGEMENT;
DOD: Department of Defense,
DOE: Department of Energy

Categorical listing	Number
Oak Ridge Associated Universities	8
DOE Contractors	7
Non-DOE Contractors	26
US Navy Submarine Programme	3
US Reactor Power Programme	2
US DOE/DOD Groups	3
Total	49

Center, University of New Mexico, Albuquerque, New Mexico, University of Washington, Seattle, Washington. With the rapid growth of organ transplantation in the USA many tertiary medical centres have excellent capabilities for care of bone marrow aplasia and its many attendant ramifications. The utilities maintaining power reactors support many of these programmes. There are several private consulting firms providing medical care and training.

Although these speciality referral centres for radiation accident victims exist, the great majority of radiation accidents involving personnel injury or contamination do not require the use of such facilities. Generally speaking, most radiation accident victims can be adequately cared for at the local hospital if the staff has received sufficient training in radiation accident management.

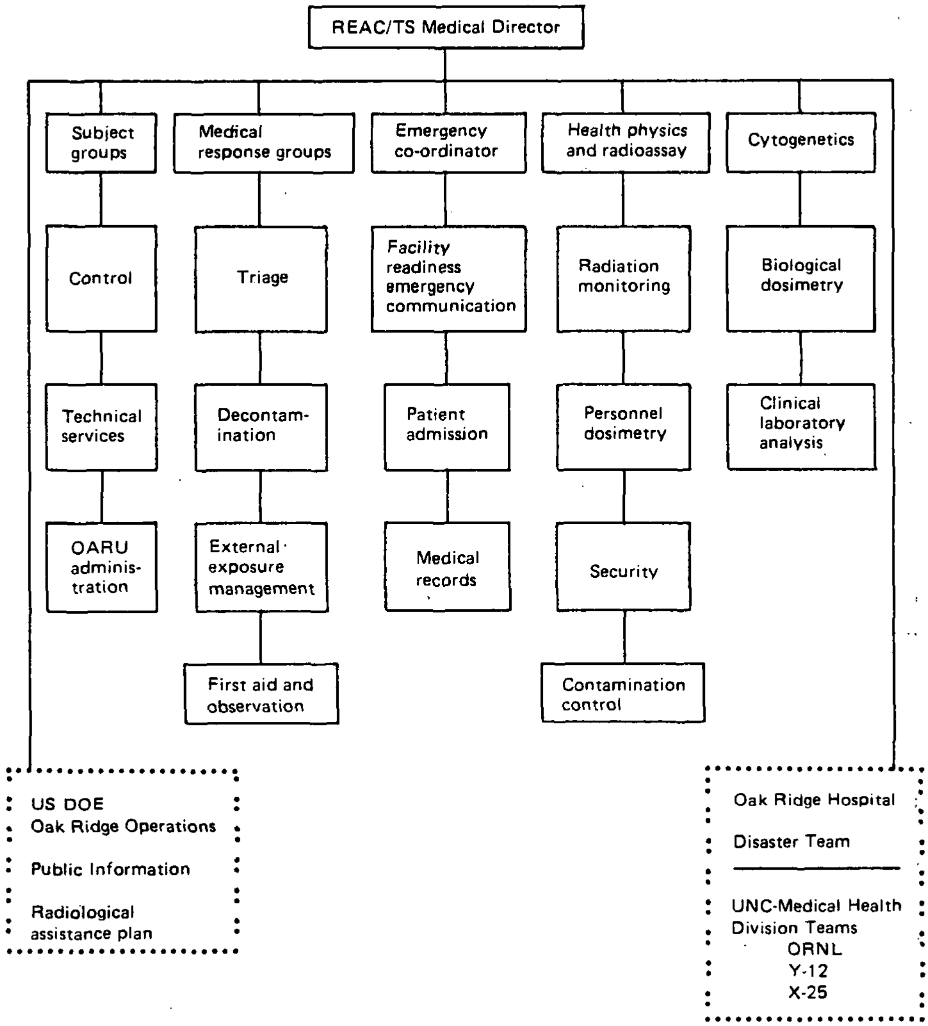

FIG. 7. Block diagram illustrating the organization of the REAC/TS emergency response team for radiation accident management; ORNL, Oak Ridge National Laboratory; UNC, University of North Carolina.

Many of these referral centres listed above have responsibility for providing planning and medical surveillance and care for accident management. Examples of REAC/TS agreements for these services are listed in Table VI. REAC/TS will also provide guidance and accept patients from elsewhere. These arrangements are made through contracts and/or letters of agreement.

REAC/TS emergency response technique

The operation of REAC/TS is comparable to that of any other emergency response facility. A radiation accident response team with adequate backup support is on 24-hour callout duty. The facility is staffed Monday through Friday, 8.00 a.m. to 4.30 p.m., and there are two remote beepers (operating through the Oak Ridge Hospital switchboard) for after-hours, weekend or holiday contact.

As with the emergency response groups noted above, REAC/TS maintains the necessary staffing required to receive, survey, and treat radiation accident victims. The staff consists of physicians, nurses, health physicists, research scientists, an emergency co-ordination and ancillary support group. A written plan for response action organizes these personnel into well defined subgroups, each with specific assignments. The radiation emergency response team concept for REAC/TS is illustrated in Fig. 7. During emergency response status, approximately 24 persons provide in-house support to the medical director. The response team can be expanded to about 50 persons when support personnel from the Oak Ridge Hospital, Union Carbide Medical Health Division teams and DOE-ORO are requested. The Oak Ridge Associated Universities (ORAU) support groups from central supply, technical maintenance and administration provide added assistance as needed.

Each aspect of emergency response to radiation accidents is practised through regularly scheduled drills. These drills, designed to convey as much realism as possible, assist each individual of the staff in identifying his role and in determining how it interrelates with the roles of others during actual radiation accident response configuration. Drills are designed to involve ancillary support groups, particularly the Oak Ridge Hospital surgery department that provides all emergency surgical support to REAC/TS.

REAC/TS training for radiation accident management

One of the most important missions of the REAC/TS programme is to disseminate information and current methodology for the medical management of radiation accidents to medical response groups remote from the Oak Ridge area. This training is vitally important to meet the needs of potential radiation accident victims, even though such accidents are quite limited in number. The primary mechanism for disseminating this information is through training courses for medical, paramedical support and health physics personnel. Indeed, the REAC/TS training programme constitutes one of the unique features of the overall programme.

Three different training activities are scheduled each fiscal year. The first course, 'Medical Planning and Care in Radiation Accidents', is designed for physicians who provide medical services to the nuclear industry, as well as city, county and state health officers, who may be called upon to provide first aid or medical care in the event of a radiation accident. The curriculum includes fundamentals of radiation and radiobiology, radiation detection and measurement, care of radioactively contaminated injuries, evaluation and treatment of internal radioactive contamination and the acute radiation syndromes. Included are demonstrations of equipment and facilities used to evaluate and treat radiation injuries. The duration of the course is five days. REAC/TS provides certification of completion of its training courses and credit for the Physicians Recognition Award of the American Medical Association. A second course, 'Health Physics in Radiation Accidents', takes also five days and is designed for health physicists who may be called upon to respond to accidents involving radioactive materials and personnel injury. The major topics covered are a review of radiation physics, principles of radiation detection and internal dosimetry, protective clothing and equipment, radiological emergency procedures and the role of the health physicist in a medical environment. Lectures are complemented by demonstrations, laboratory exercises and a simulated radiation accident drill. This course is approved by the American Board of Health Physics for continuing education units and recertification of health physics personnel. A third course, 'Handling of Radiation Accidents by Emergency Personnel', is of 2.5 days duration and is approved by the American College of Emergency Physicians and the Tennessee Nurses Association for continuing education credit. The course is designed primarily for emergency room physicians and nurses who may be called upon to administer initial hospital aid to a radiation accident victim. The course emphasizes the practical aspects of handling a contaminated victim and deals with the fundamentals of radiation, how to detect and measure it, how to prevent the spread of contamination, how to reduce the radiation dose to the victim and attending personnel, and how the medical physicist treats contaminated accident victims. Lectures are complemented by demonstrations, laboratory exercises and a simulated accident drill.

In each of these three training endeavours, participation is limited, thereby providing for optimum student–faculty ratios and the opportunity for everyone to participate in laboratory and drill activities. The teaching faculty serving these training courses includes both REAC/TS personnel and nationally recognized experts in radiation accident management. By background training, the faculty includes MDs, PhDs, certified health physicists, registered nurses, licensed practical nurses, as well as other professional biologists, chemists and physics personnel.

Laboratory exercises and accident drills are designed to provide meaningful, realistic information and the opportunity to practise techniques for emergency

TABLE VII. SUMMARY OF REAC/TS TRAINING COURSE ACTIVITIES[a] COVERING THE PERIOD OF FACILITY OPERATION FROM ITS BEGINNING IN 1976 TO THE END OF THE FISCAL YEAR 1987; EMT: Emergency Medical Technologists; HP: Health Physicists; ADM: Administrative Staff

Fiscal year	MD	PhD	Nurses	ADM	EMT	HP	Other	Total
1976	14	—	2	—	—	2	1	19
1977	41	3	22	8	2	18	4	98
1978	49	3	25	6	4	23	17	127
1979	59	6	30	—	5	34	7	141
1980	66	4	45	2	—	30	22	169
1981	89	6	53	2	—	29	16	195
1982	79	4	55	2	1	23	26	190
1983	74	8	63	2	2	29	18	196
1984	68	1	57	—	2	41	17	186
1985	73	—	52	—	1	36	13	175
1986	62	3	60	2	3	39	9	178
1987	92	3	64	—	3	72	40	274
Total:	766	41	528	24	23	376	190	1948[b]

[a] Includes courses: Medical Planning and Care in Radiation Accidents, Health Physics in Radiation Accidents, and Handling of Radiation Accidents by Emergency Personnel.
[b] Includes participants from 16 foreign countries.

response to radiation accidents. Course participants complete written examinations and critique forms at the close of each respective course. Critique summaries and test scores are mailed to all former course participants. The REAC/TS training course activities are summarized in Table VII.

2. CIR — CENTRE INTERNATIONAL DE RADIOPATHOLOGIE, INSTITUT CURIE — PARIS, FRANCE

The CIR was created by the Institut Curie and the Commissariat à l'énergie atomique (French Atomic Energy Commission). The World Health Organization has designated CIR as an International Collaborating Centre.

Action procedure in case of radiological accident

In case of a radiological accident involving persons, whether related to external or internal exposure, the International Centre of Radiopathology (CIR) is equipped and organized to answer calls, and to mobilize and activate the medical and technical teams promptly. The Centre is operational 24 hours a day, seven days a week. The means developed cover two different aspects:

(i) The Centre can be alerted by an outside body, whatever the day or time, to obtain the first information required to take action. The dialogue between the Centre and the applicant is indispensable and must be established rapidly and efficiently. This arrangement constitutes the Outside Procedure.
(ii) Once the Centre has identified the call and the type of action required, and hence the type and size of the specialized teams to be activated, it must be able to mobilize its own resources rapidly. This arrangement constitutes the Internal Procedure.

Outside procedure

The outside procedure actually includes two different aspects:

(i) The first essentially concerns communications and answers the question: How can one discuss the matter rapidly with the Centre?
(ii) The second concerns information of a medical, biological and dosimetric nature, etc., that the applicant can already provide to the Centre, and which may be of primary importance for diagnosis, therapy and prognosis.

Communications

The Centre can be alerted in many ways: by telephone, telegram, telex or telecopy.

Telephone

The Centre has two international lines:

(1) 654 49 29
(1) 654 49 30

preceded by 33 for calls coming from other countries and by 16 from France.

During working hours, the phone call is received by an operator who has a short list of questions designed primarily to identify the applicant and the type of accident, and who immediately connects the call to a responsible physician at the Centre.

Outside working hours (nights, weekends, holidays), the caller is connected to an answering machine that asks a number of questions in French and English. Simultaneously, the call is automatically switched to the Operational Safety Station of the Fontenay-aux-Roses Nuclear Research Centre. This station immediately alerts a physician of the Centre, to whom it communicates all the data recorded on tape. This physician can therefore speedily enter into direct contact with the applicant. The following text is recorded on tape.

"This is the International Radiopathology Centre. Your call is being recorded.
— Please state your name, address and phone number.
— Details concerning the accident, irradiation or contamination.
— Transport facilities."

In addition, the Safety Control Section can communicate directly with the caller and record the initial data for transmission to the responsible physician.

Telex

The Centre has a telex line: ENERGAT 270898

This telex can be used to broadcast the alert. It offers the advantage of leaving a trace, of avoiding any error of interpretation that may be due to difficulties of a linguistic nature, and of communicating more precise data than is feasible by telephone. It has the drawback of not allowing direct dialogue immediately. To avoid any error or false interpretation, it is advisable to use this telex facility, either to back up the emergency phone call, or independently. It has always proved indispensable for additional information to be transmitted to the Centre, either before or after it has been alerted.

Telecopy machine

The Centre has a telecopy machine:

(1) 654.46.10

preceded by 31 for calls coming from other countries and by 16 for calls from France.

This telecopy machine obviously cannot be used to alert the Centre, but to transmit documents that may be vitally important, such as drawings of the premises and the position of the victims, to allow a recreation of the incident, dosimetric calculations, haematological curves, etc. In case of linguistic or administrative difficulties, it can be used to transmit texts that facilitate certain procedures and eliminate these difficulties.

Information required

The information to be furnished covers various aspects, both administrative and medical.

Administrative and general information

This information helps to find the best solution to the problems of transport of the victim, and to mobilize the competent team which will thus be ready to receive him. The Centre essentially needs to know the following exactly:

Activity of the applicant

(a) Identity of the caller
(b) Position
(c) Identification of the establishment
(d) Address
(e) Phone number, telex

Circumstances of the accident

(f) Date
(g) Time
(h) Type
(i) Nature: — Irradiation, source
— Contamination, radionuclide(s) involved
(j) Associated injuries or none, localization
(k) Risk of contamination for the environment
(l) Action desired: — Dispatching of a team
— Transport to the CIR

Transport

(m) Date, time and place of departure
(n) Date, time and destination
(o) Type of transport planned:
— Air: — flight number and airline
— departure airport
— destination airport
— arrival time
— Rail: — train number
— departure station
— destination station
— arrival time

— Car: — number plate
— place of departure and arrival
— arrival time.

(p) Confirmation of time-tables, arrival times and flight or train number
(q) Number of victims
(r) Identity of victims
(s) Number of accompanying persons
(t) Identity and status of accompanying persons
(u) Need for stretchers, if any
(v) Need for an ambulance, if any
(w) Volume and type of luggage or equipment accompanying the victims
(x) Need, if any, for special packs for specimens, if required.

Medical and biological data

These data are indispensable and serve to hospitalize the victim in the best possible conditions. They guide the diagnosis, may indicate additional examinations to be performed urgently, and make it possible to begin the treatment. The best way to proceed is to compile these data on a standard Accident Form. This form is designed to:

— Gather together all the initial dosimetric and clinical data concerning the person suspected of external irradiation;
— Guarantee liaison between the specialized services which will take charge of the victim subsequently.

It also serves to review the tests, examinations and specimens taken in this case.

It is made up of several parts. If the accident occurred in a sufficiently large installation having specialized services permanently installed on the site (such as Prevention and Radiological Service, a Medical Service, a Biological Analysis Laboratory), the different parts of the form will go to these different services. In addition, a sheet will follow the victim in case of transport to the specialized medical service.

A sample irradiation Accident Form is given in Appendix 3.

General organization and assistance agreement

For example, it may consist in:

— a detailed agreement mentioning only the medical intervention in case of an accident,

— a simpler agreement mentioning only the medical intervention in case of an accident,
— a very general agreement, covering various fields, such as clinical and/or experimental research,
— an agreement mentioning only specific points,
— a simple convention for specific actions.

Internal procedure

As soon as the operator receives the alert, a responsible physician is contacted as speedily as possible. During working hours, the procedure is extremely simple, and the caller may be connected directly to the physician so that the dialogue can be established immediately. Outside working hours, the Operational Safety Station has the means necessary to alert one or more physicians. For this purpose, it has a 'roster' of ten physicians who can be mobilized, including for each of them:

— Home phone number,
— Phone number of country and weekend residence,
— Call number of the personalized Eurosignal receivers designed to contact a person who cannot be called by telephone: five physicians are in permanent possession of a Eurosignal receiver covering France and the Federal Republic of Germany.

The 'rosters' are updated, checked regularly, and temporary exclusions are indicated to the Operational Safety Station. In addition, each physician on the 'roster' has the list of specialists who must be mobilized rapidly. This list includes the normal, weekend and holiday addresses as well as the phone numbers. Two Eurosignal receivers are allocated to the managers of the two laboratories who may be required to intervene in emergencies, for haematology/cytogenetics and dosimetry. These specialists work in the following areas:

— Biology
— Dosimetry
— Radiotoxicology
— Detriment.

These twelve first aid specialists have the possibilities of contacting, outside working hours and for follow-up treatment, the technicians concerned in the case of an accident requiring the rapid activation of large scale resources.

Appendix 3

SPECIMEN OF ACCIDENT REPORT FORM

Form No. 1

Part to be kept at the advanced medical unit

Family name: First name: Date:
Sex: M F Identification No.:

Evacuated to: at: h

1. FIRST AID ADVANCED MEDICAL UNIT

Date: h:
Family name: First name: Age:
Nationality: Sex: M F
Particular signs:

Lesions	crane	face	neck	rachis	thorax	abdomen
fracture						
contusion						
wound						
burn						

CLINICAL STATE
Blood pressure:
Pulse:
Consciousness:
Ventilation:

DIAGNOSIS
by Dr.:

TREATMENT h:

PRIORITY FOR EVACUATION
Absolute emergency:
Secondary emergency:
death:

EVACUATION: lying: seated: standing:
Means of evacuation:
to: (medical triage centre, hospital, etc.)

2. MEDICAL EVACUATION CENTRE

Precise diagnosis after triage	Blood pressure: Pulse: Consciousness: Ventilation:

TREATMENT	h:

PRIORITY FOR EVACUATION — death:

- Absolute emergency
  - priority
  - urgency 1
- secondary emergency
  - urgency 2
  - urgency 3

EMBARKING POINT

- by air: helicopter / plane
- by road: intensive care ambulance / ambulance / car
- by train

lying: medically accompagnied yes / no — seated — standing

DESTINATION:

TRANSPORTATION

Departure: day: h

Clinical evolution, treatment:

Part to be kept at the embarking point:

Name:	No:
First name:	
Address:	
Sex: M F	EVACUATED
Coming from:	on the: at: h
(Advanced medical unit)	by:
	to:

Form No. 2

IDENTIFICATION MEDICAL FORM
In case of accidental irradiation
Identification No. (auto stickers)

IDENTIFICATION OF THE VICTIM:
Name: (upper case)
First name:
Birthday: / / / / / / /   Sex: M F

IDENTIFICATION OF THE INDIVIDUAL WHO FILLS THE FORM:
Name:
Function:
Organization:

Date and hour of FORM FILLING
Date: / / / / / /   Hour: / / / / /

ACCIDENT:
Date of exposure: / / / / / /   Presumed hour: / / / / /

EXPOSURE CONDITIONS:
DURATION: / / / /
If possible, time beginning: / / / /
end: / / / /
Position of the victim:
Occupation of the victim:

DOSIMETRY:
The victim had a dosimeter: yes no
Dosimeter recovered: yes no
if yes: No. / / / / / / / / / /

RESPIRATORY PROTECTION: yes no

CONTAMINATION OF CLOTHES (if detected)
yes no

FIRST SYMPTOMS

CLINICAL STATE OF THE VICTIM

	time of appearance	Number or duration
Nausea	/ / / / /	
Vomiting		
Wound		
Trauma		
Burn		

MEDICAL FINDINGS (to be filled by the physician)

Name of physician: (upper case)
Name of victim: (upper case)
First name:

Date of examination: / / / / / / /   Hour: / / / / / /

asthenia:	yes	no		
cephalea:	yes	no		
			time of appearance	
nausea:	yes	no	/ / / / / / /	duration:
vomiting:	yes	no		number:
diarrhoea:	yes	no		quantity:

temperature: / / / / / /
pulse:
blood pressure:
consciousness: normal   abnormal: agitation
delirium
sleepiness
coma

equilibrium perturbation:	yes	no
coordination perturbation:	yes	no
skin and mucosa: oedema	yes	no
erythema	yes	no

other:

MEASURES TAKEN

Undressing:	yes	no
Decontamination	yes	no
DTPA:	yes	no

if yes, administration pathway:
- aerosol
- bathing
- intravenous

Stable iodine:	yes	no

LABORATORY TESTS

BLOOD SAMPLES

* First sample (if possible, before the third hour)

  Date: / / / / / / Hour: / / / / /

  - Blood cell count, platelets: yes no
  - Cytogenetic (10 ml): yes no
  - Sample for spectrometry: yes no

* Second sample (if possible 2 hours after the first one)

  Hour: / / / / /

  - Blood cell count, platelets: yes no
  - HLA typing: yes no

URINE SAMPLES: yes no

(if possible for gamma spectrometry)

Is it the first urination after the accident? yes no

PHYSICIAN CONCLUSIONS:

DESTINATION OF THE VICTIM:

Appendix 4

EARLY CLINICAL SYMPTOMS ASSOCIATED WITH PARTIAL BODY EXPOSURE

EARLY CLINICAL SYMPTOMS ASSOCIATED WITH PARTIAL BODY EXPOSURE
(Chernobyl cases, contributed by A. Baranov and A. Barabanova)

Time post-irradiation	Dose and diagnostic signs of damage due to partial body exposure					Notes
	1. Common symptoms irrespective of the part of body irradiated	2. Extremities	3. Abdomen	4. Breast	5. Head	
24 h	1.1. Leucocytes, over 20×10^3 cells per 1 μL	2.1. Primary erythema appeared within 5 h, increased later and was accompanied by oedema	3.1, 4.1, 5.1. Non-conformity between pronounced primary response and its delayed onset (e.g. uncontrollable vomiting started within 3 h after exposure)			Pronounced (3.3) and/or (5.4) indicate extremely severe damage with early fatal outcome
	1.2. Lymphopenia, less than 0.5×10^3 cells per 1 μL		3.2. Primary abdominal skin erythema appeared within 10–12 h	4.2. Primary breast skin erythema appeared within 10–12	5.2. Primary face erythema appeared within 3–4 h and hyperaemia of conjunctives	

	Dose and diagnostic signs of damage due to partial body exposure					
	1.3. Body temperature above 38°C		3.3. Diarrhoea	4.3. Tachycardia (over 100 beats per min), arterial hypotension, changes on electrocardiogram (ECG)	5.3. Changes in radioencephalogram (REG) and electro-encephalogram (EEG) 5.4. Disturbance of consciousness	
72 h	1.4. Lymphopenia, 0.2×10^3 cells per 1 μL or less	2.2. Erythema and oedema retained or increased 2.3. Blisters	3.4. Recurrent vomiting, lack of appetite 3.5. Persistent abdominal skin erythema	4.4. Steady or increased (4.2) and (4.3) (yet their disappearance or reduction does not imply mild case)	5.5. Prolonged primary response (headache, general weakness, transient changes in EEG and REG 5.6. Persistent face erythema, and hyperaemia of conjunctives	Pronounced (1.4) (lymphocytes, 0.0) and (2.3) indicate threat to life and that of early amputation, respectively

EARLY CLINICAL SYMPTOMS ASSOCIATED WITH PARTIAL BODY EXPOSURE
(Chernobyl cases, contributed by A. Baranov and A. Barabanova)

	Dose and diagnostic signs of damage due to partial body exposure					
8d	1.5 Persistent lymphopenia 1.6. Neutropenia, less than 1.0×10^3 cells per 1 μL	2.4. Pronounced secondary erythema, oedema and blisters	3.6. Discomfort, abdominal pain, atonic constipation or diarrhoea (erythema may be absent) 3.7. Secondary erythema, oedema, blisters, etc.	4.5. Tachycardia (over 100 beats per min), arterial hypertension, changes on ECG	5.7 Oropharyngeal syndrome with ulcerous alterations, signs of eye damage	Symptom (3.7) indicates extremely severe intestinal damage

Appendix 5

DEFINITIONS

This appendix gives a selection of words and terms used in this publication with their meaning as defined in IAEA Safety Series No. 76.

abnormal exposure conditions — Conditions in which a source or the radiation from it is not under control.

accident — In the context of nuclear safety or radiation protection, an event which leads or could lead to abnormal exposure conditions.

acute exposure — Duration of the irradiation does not extend beyond a few hours.

contamination, radioactive — The presence of a radioactive substance or substances in or on a material or in a place where they are undesirable or could be harmful.

decontamination — The removal of radioactive contaminants with the objective of reducing the residual radioactivity level in or on materials, persons or the environment.

deposition — Amount of radioactive material incorporated into tissues and organs (see **intake** and **uptake**). Also used to denote the process.

external exposure — Irradiation by sources outside the body.

inhalation — Intake of material by way of the respiratory system (including the material which will eventually go to the intestinal system).

intake, radioactive nuclide	Amount of radioactive material introduced into the body by inhalation or ingestion, or through the skin (see also **uptake** and **deposition**). Also used to denote the process.
internal exposure	Irradiation by sources inside the body.
localized irradiation	Irradiation of a localized small area of the body, usually the extremities (hands, forearms) resulting from exposure to a narrow beam or holding a source, e.g. in the hand. Localized irradiation, however severe, does not lead to the occurrence of the acute radiation syndrome.
non-stochastic	Radiation effects for which a threshold exists above which the severity of the effect varies with the dose.
partial irradiation	Irradiation of a significant part of the body (chest, stomach, abdomen, half of body). Could (from the clinical point of view) lead to whole body irradiation syndrome (more or less important according to bone marrow affected, e.g. chest).
sealed source	A source whose structure is such as to prevent, under normal conditions of use, any dispersion of the radioactive material into the environment.
stochastic	Radiation effects, the severity of which is independent of dose and the probability of which is assumed by the ICRP to be proportional to the dose without threshold at the low doses of interest in radiation protection.

uptake	Amount of radioactive material absorbed into the extracellular fluids (see **intake** and **deposition**). Also used to denote the process.
whole body exposure	Exposure resulting from irradiation (usually acute) of the whole body or of most of the body volume.

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and Treatment of Overexposed Individuals
IAEA, Vienna, 6–10 October 1986**

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**Technical Committee Meeting on
General Principles and Recent Advances in
the Diagnosis, Prognosis and Treatment of
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Paris, 27–31 October 1987**

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